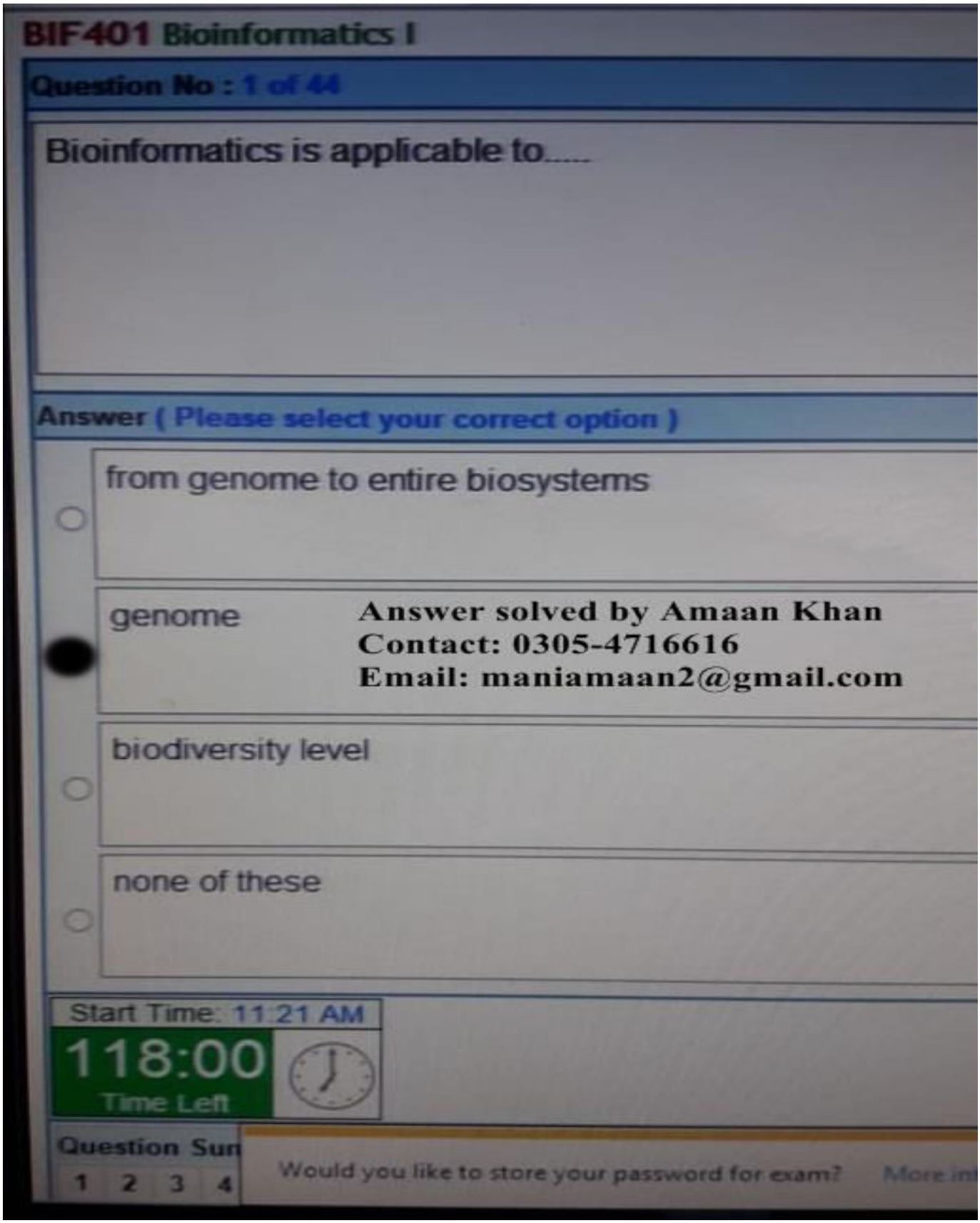
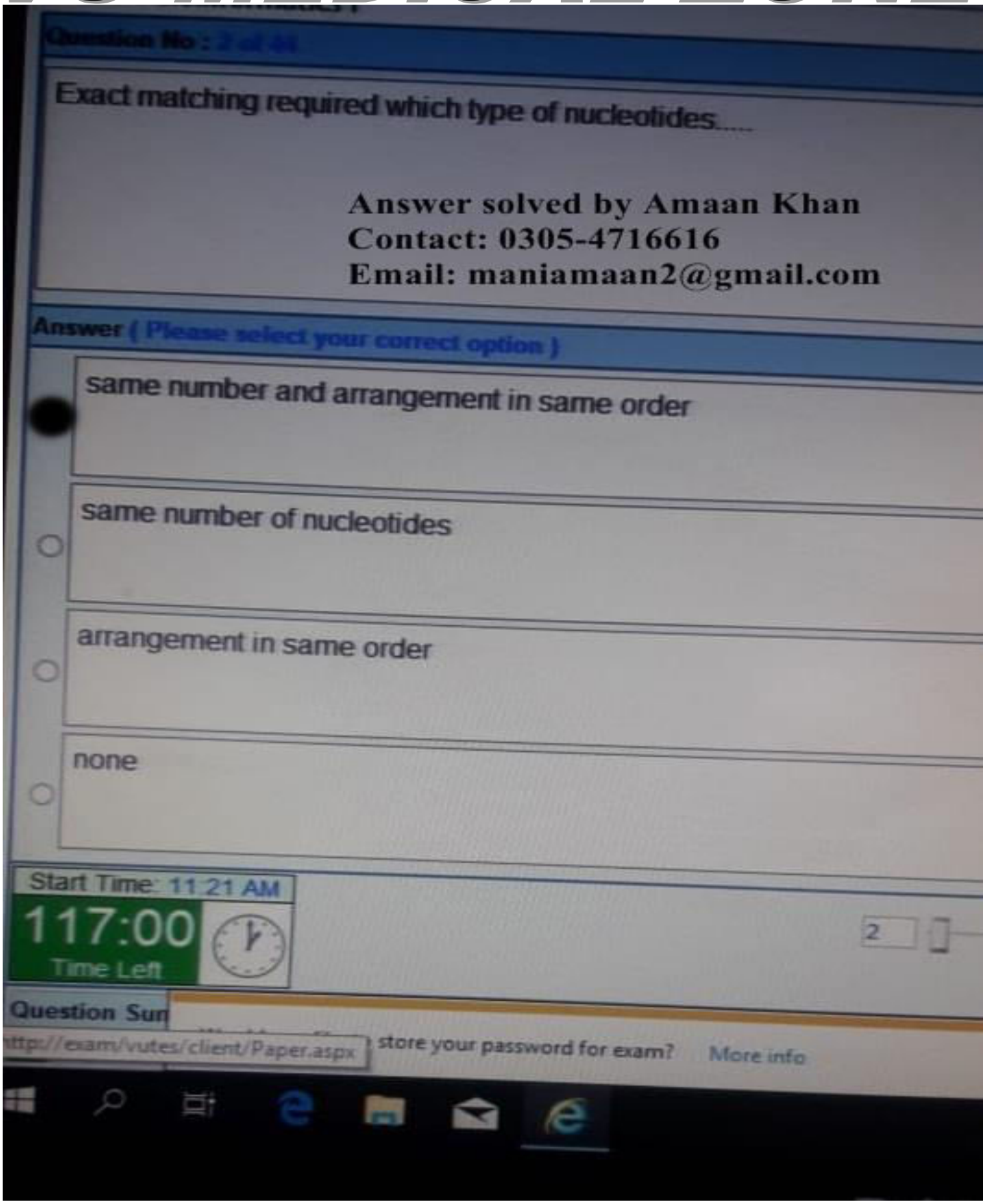
VU MEDICAL ZONE

BIF 401 Current Final Terms solved



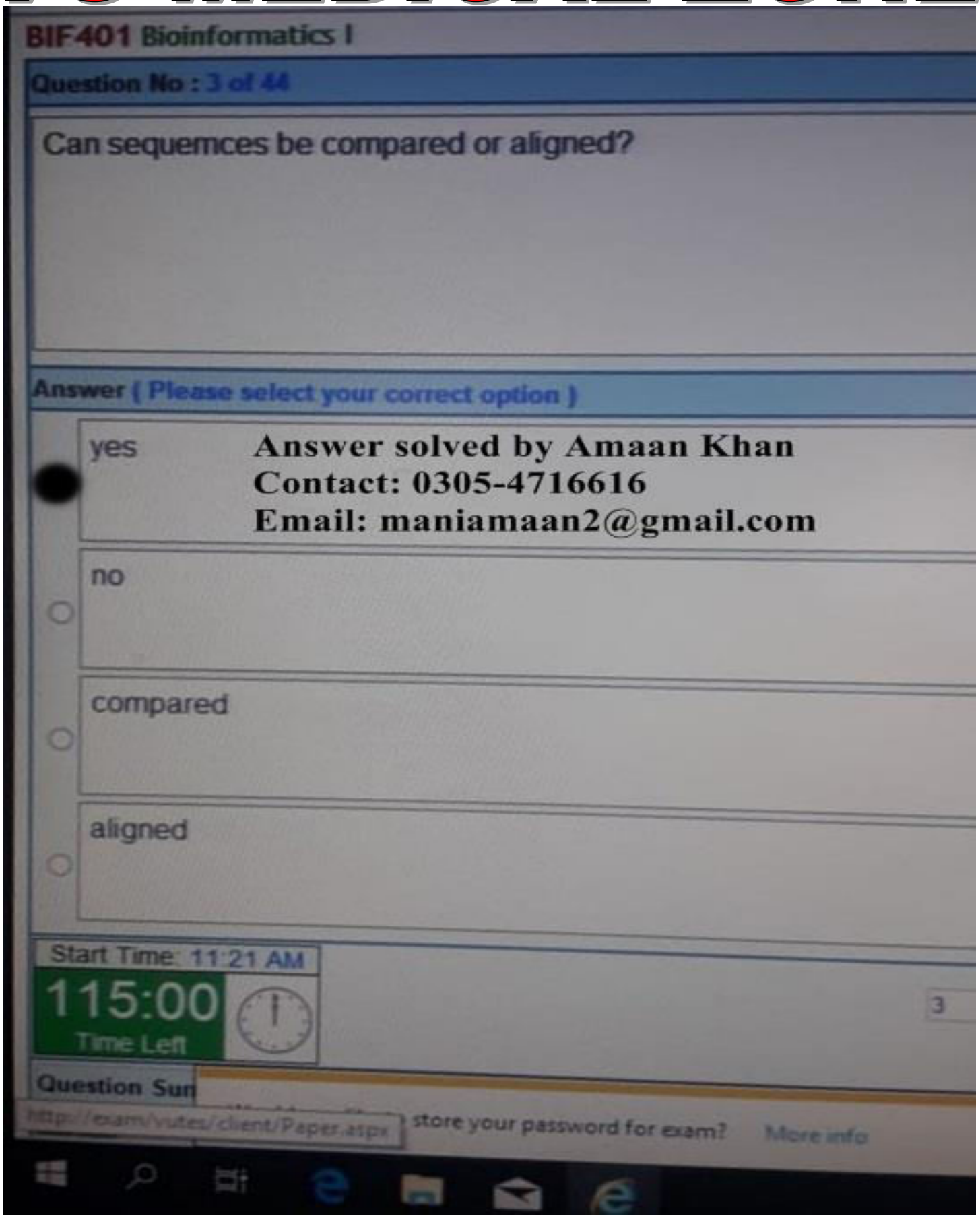
EFFORT BY AMAAN KHAN

VU MEDICAL ZONE



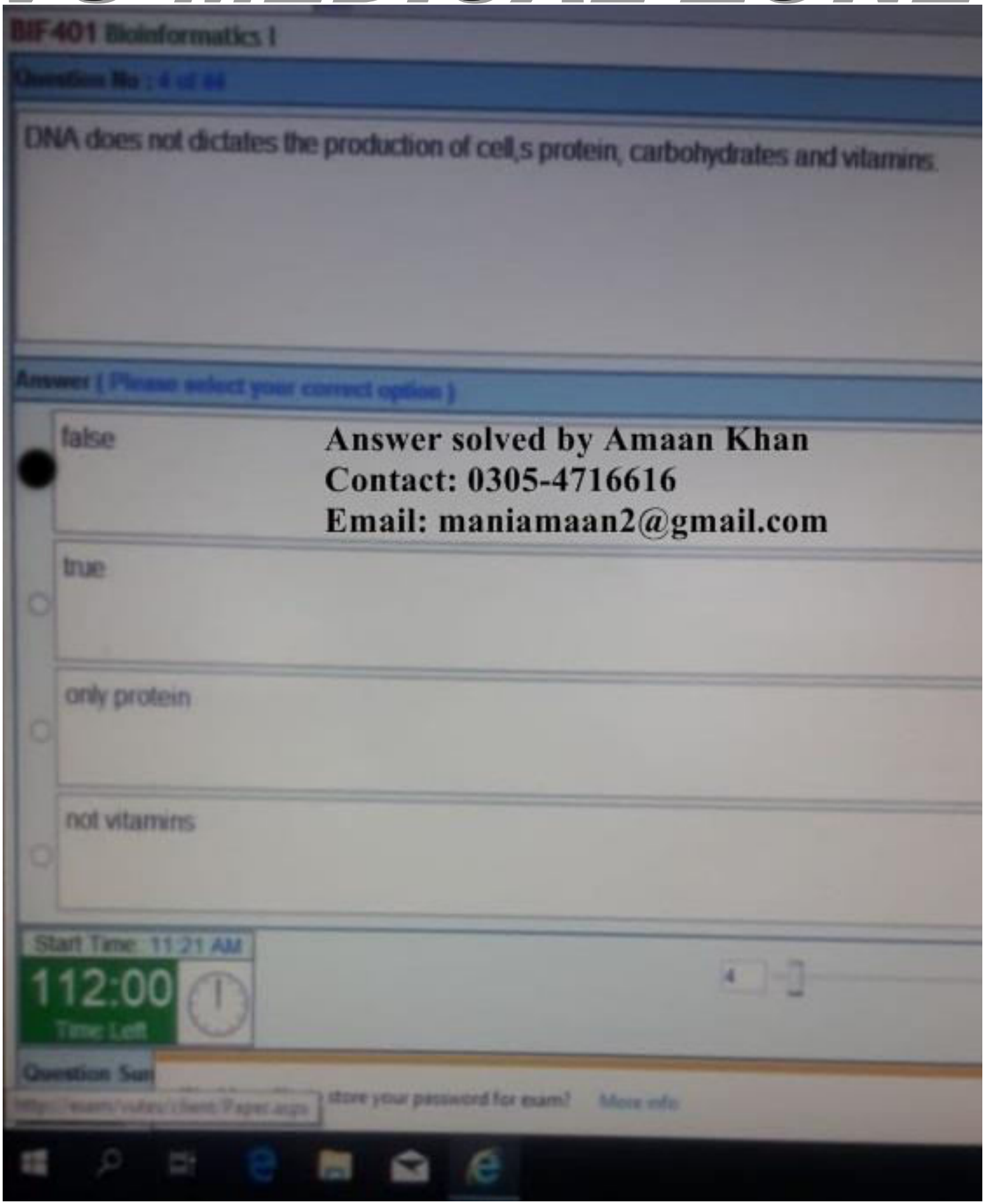
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VU MEDICAL ZONE



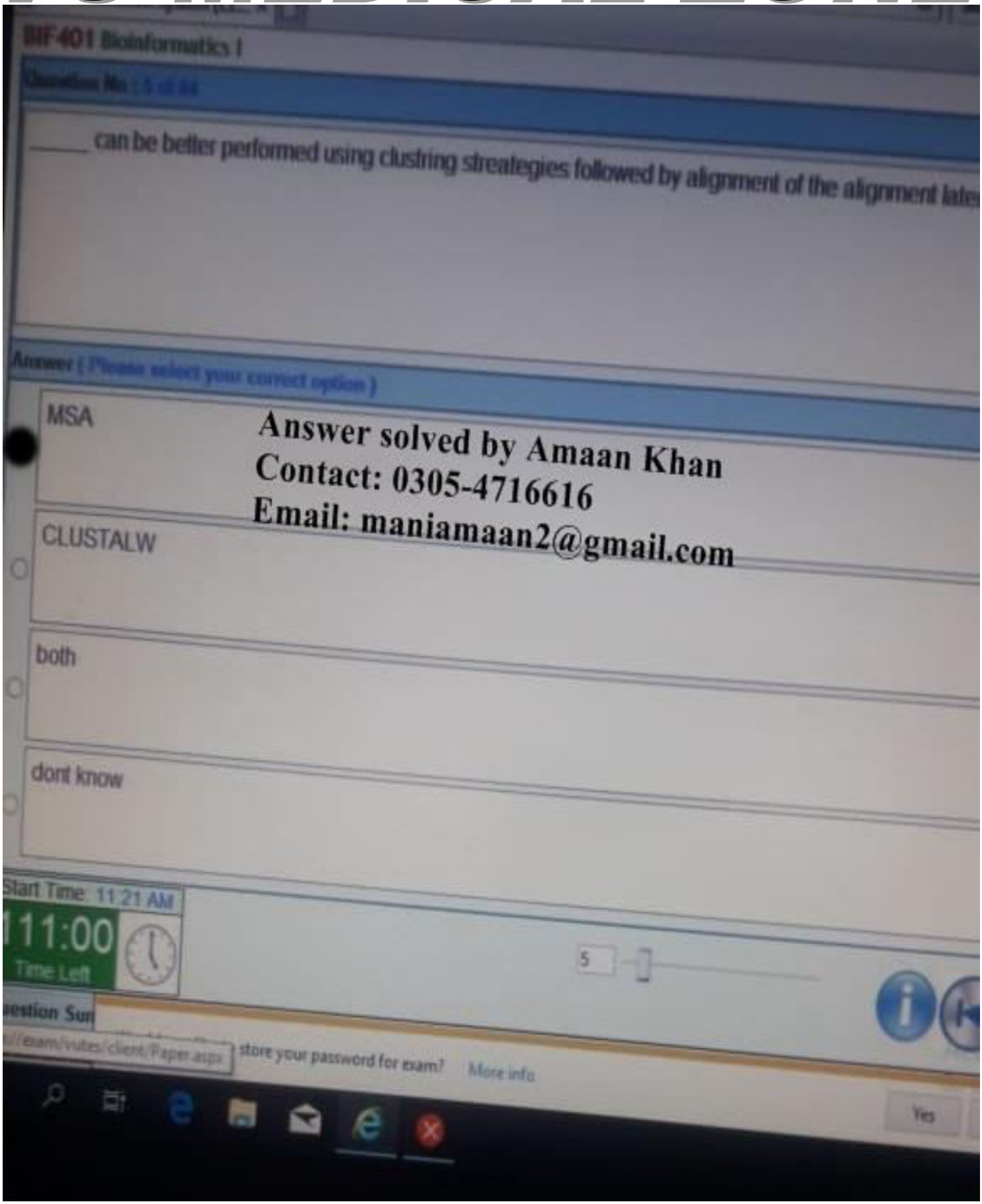
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VU MEDICAL ZONE



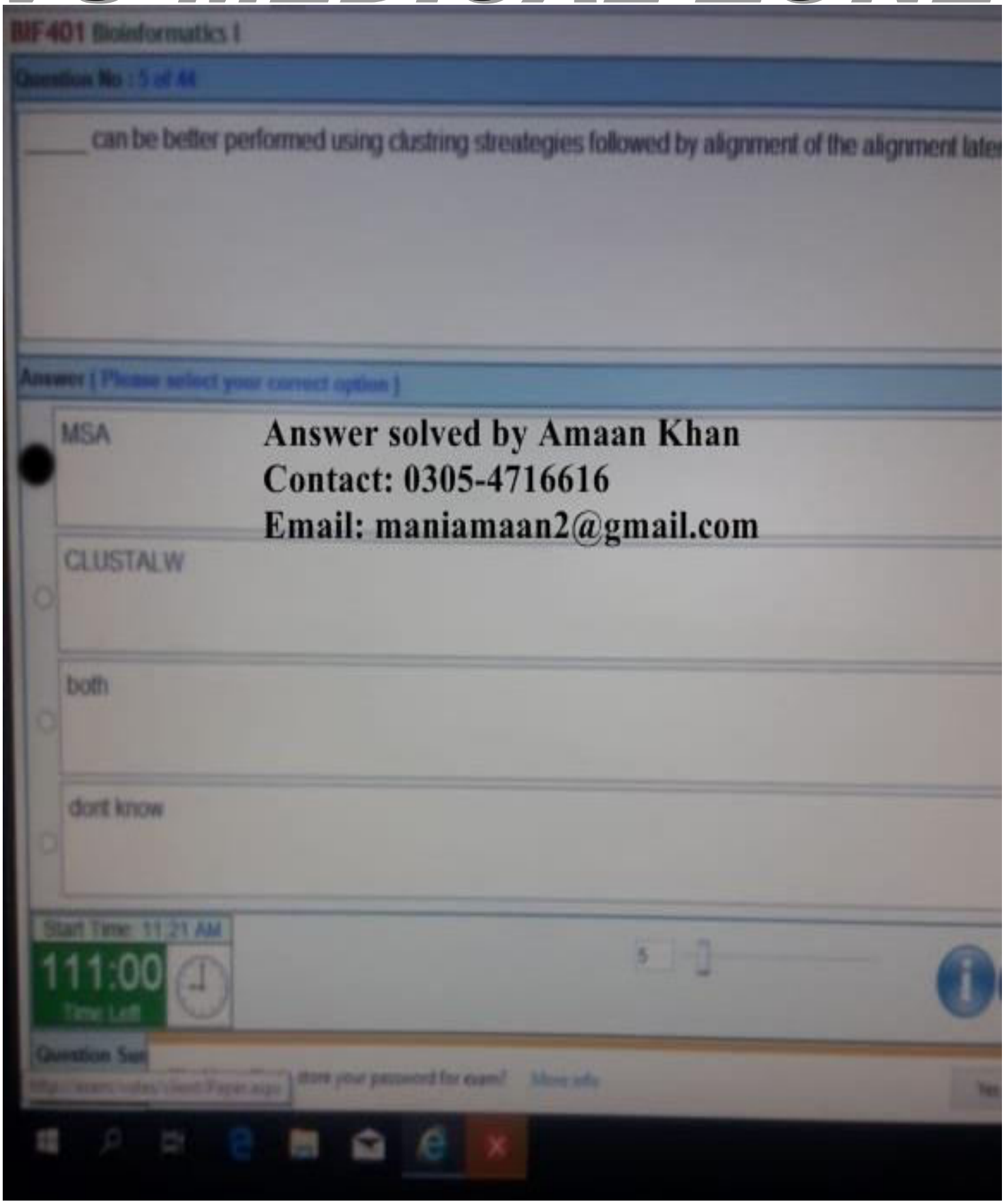
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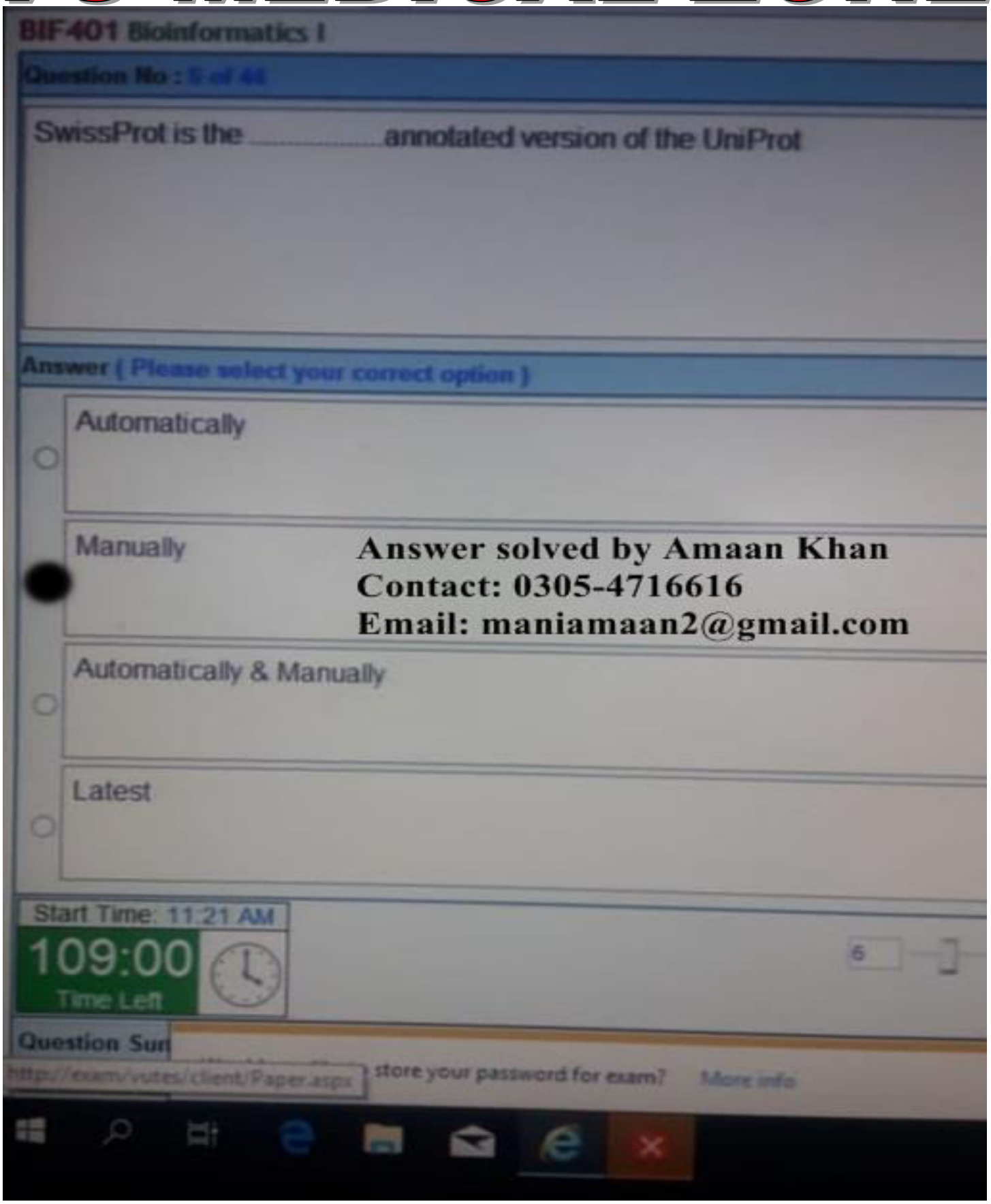
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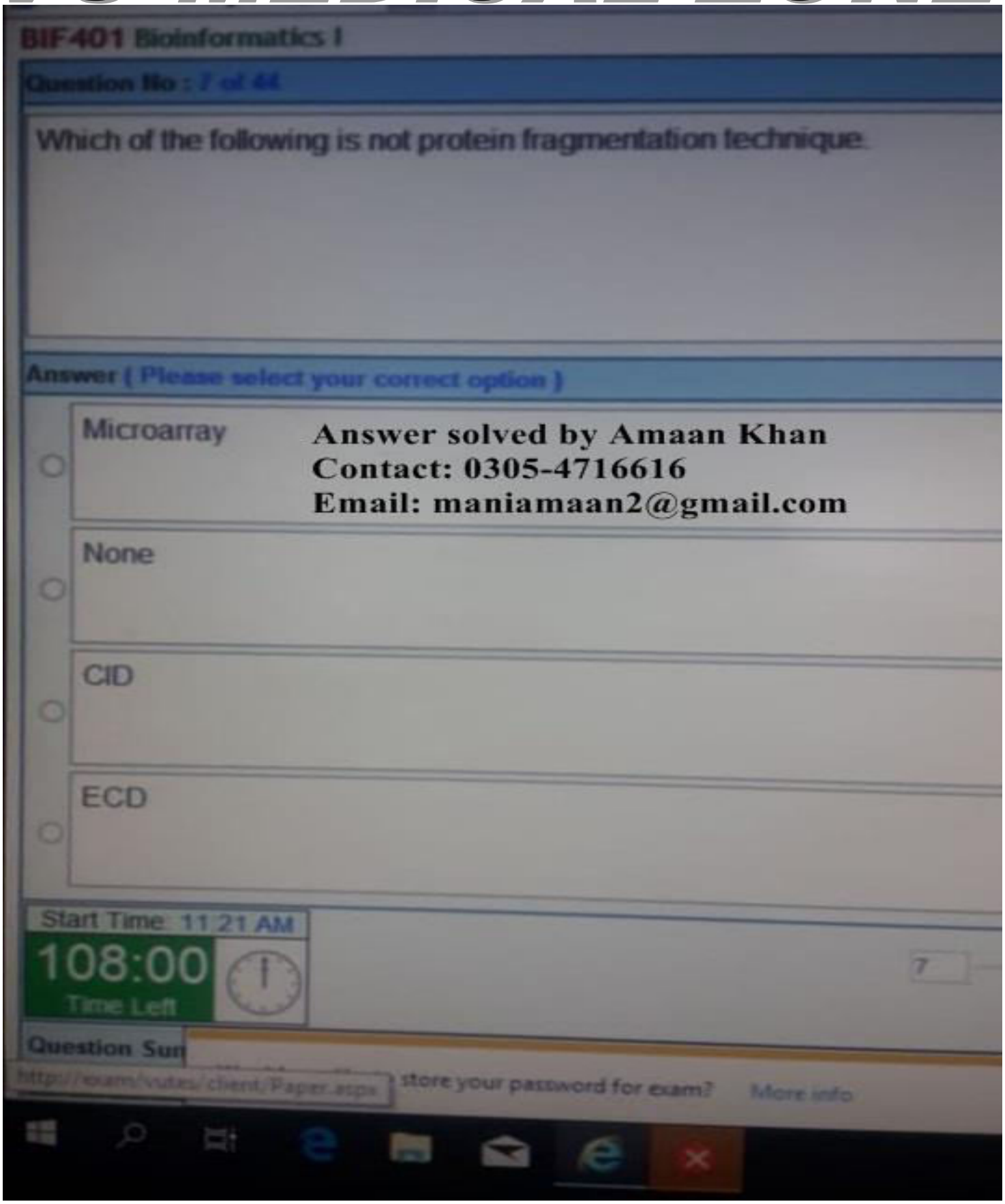
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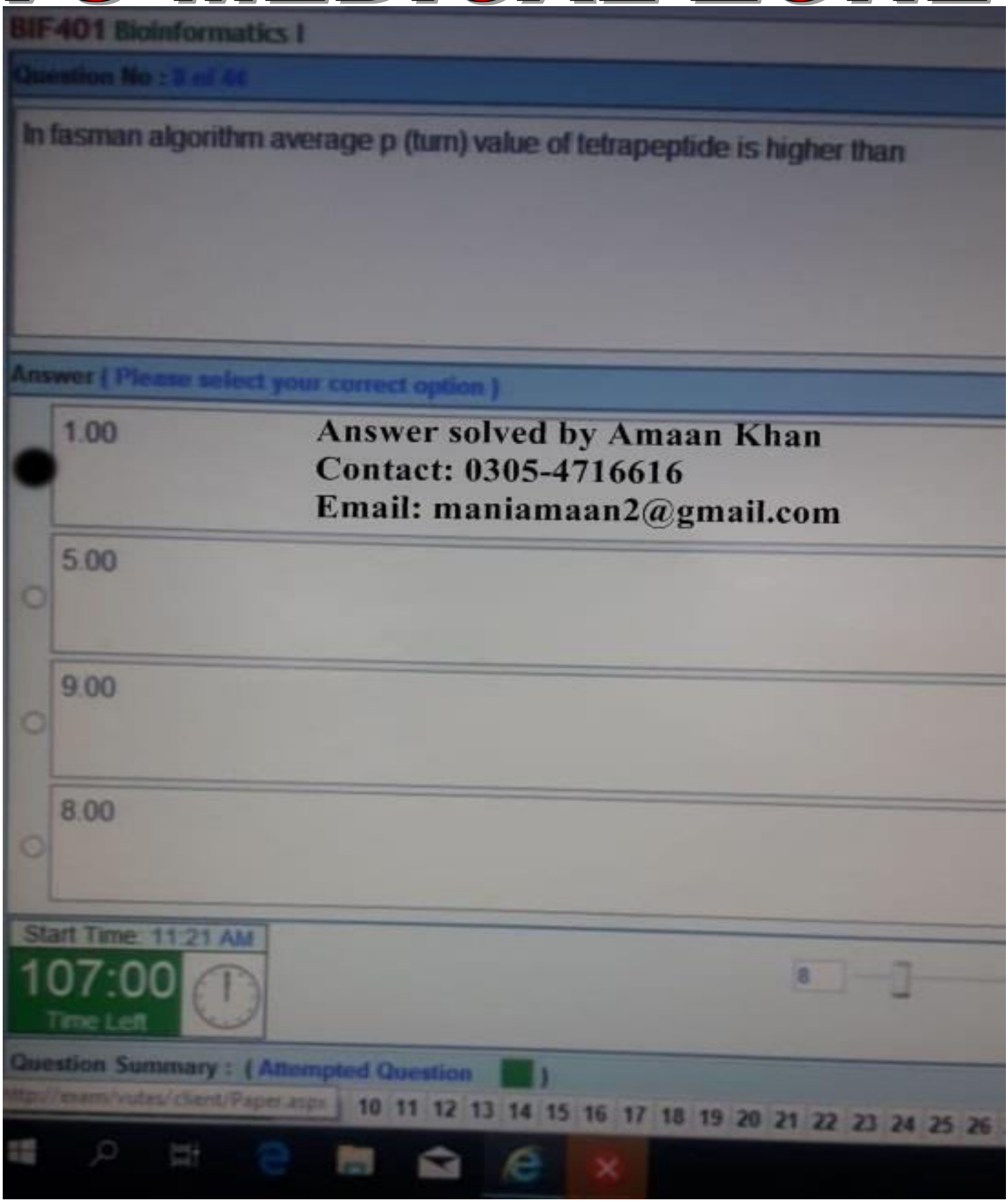
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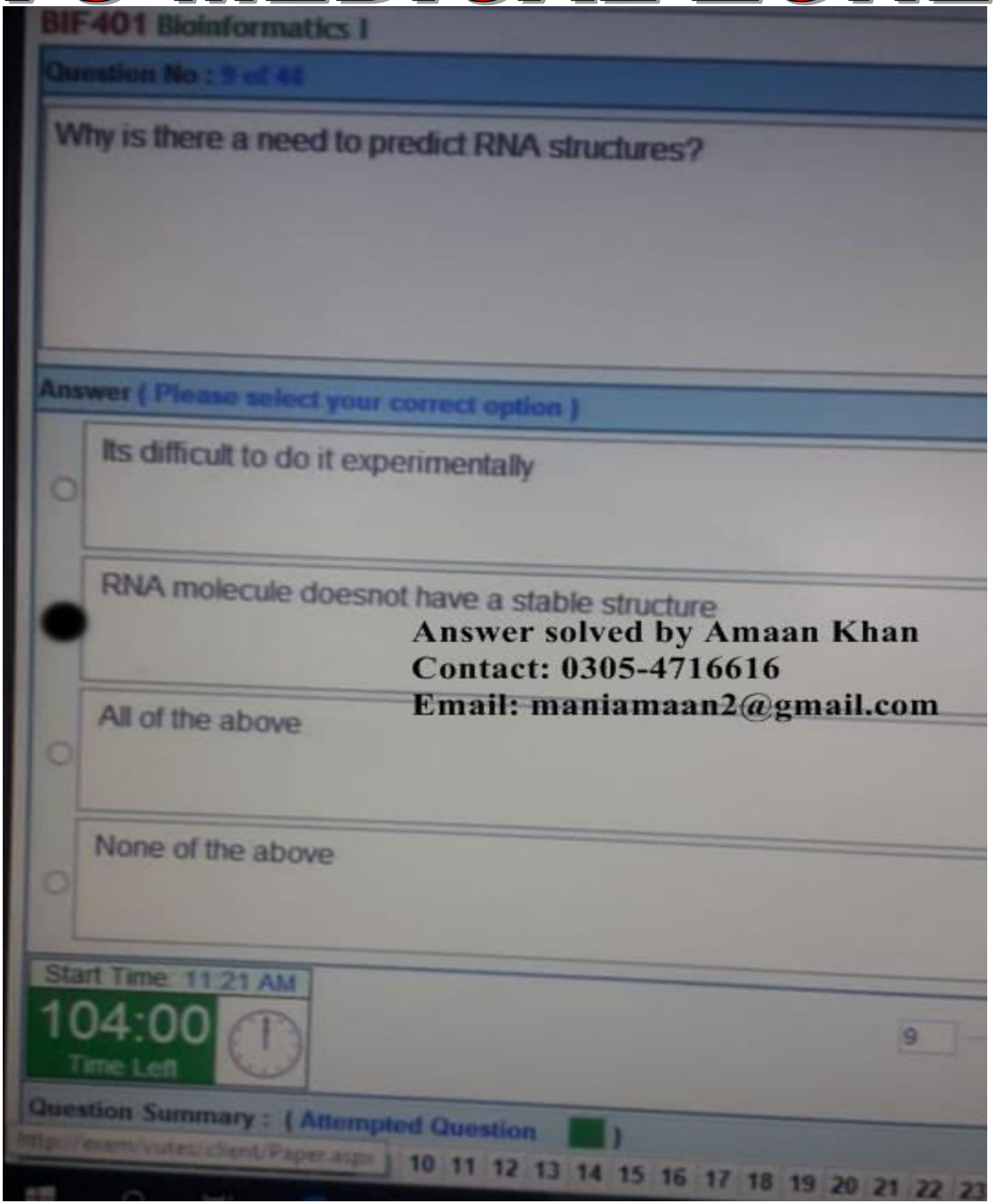
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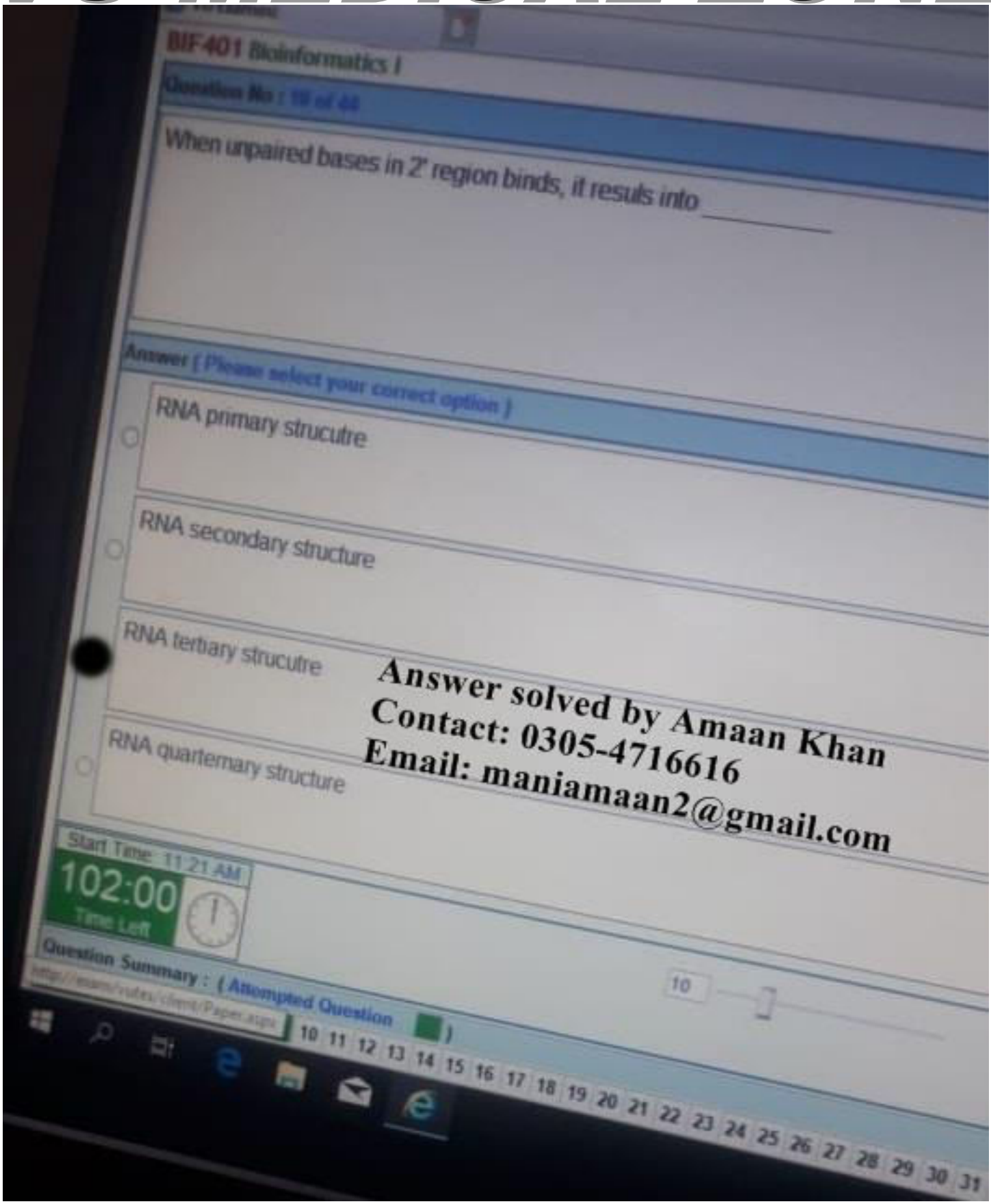
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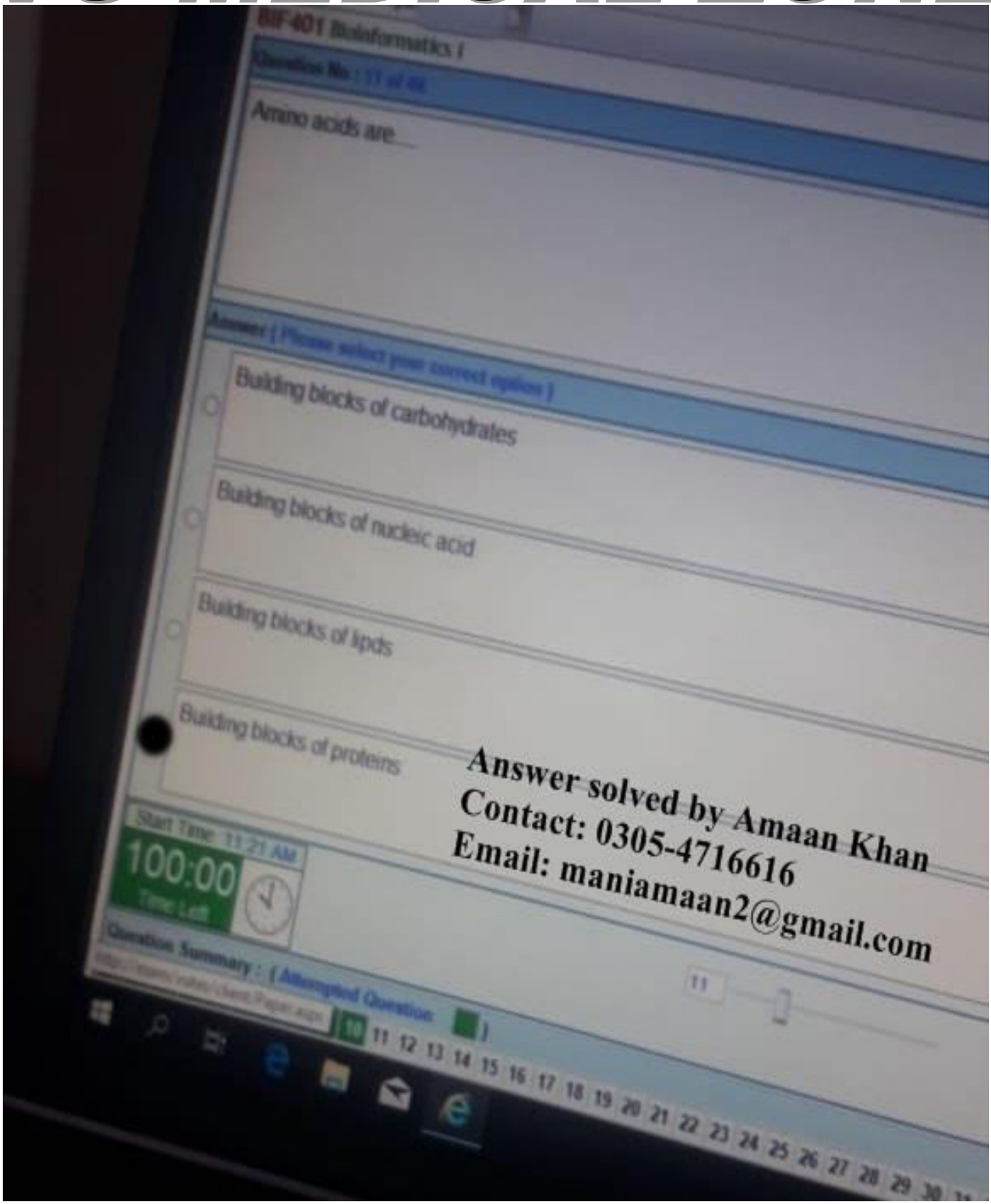
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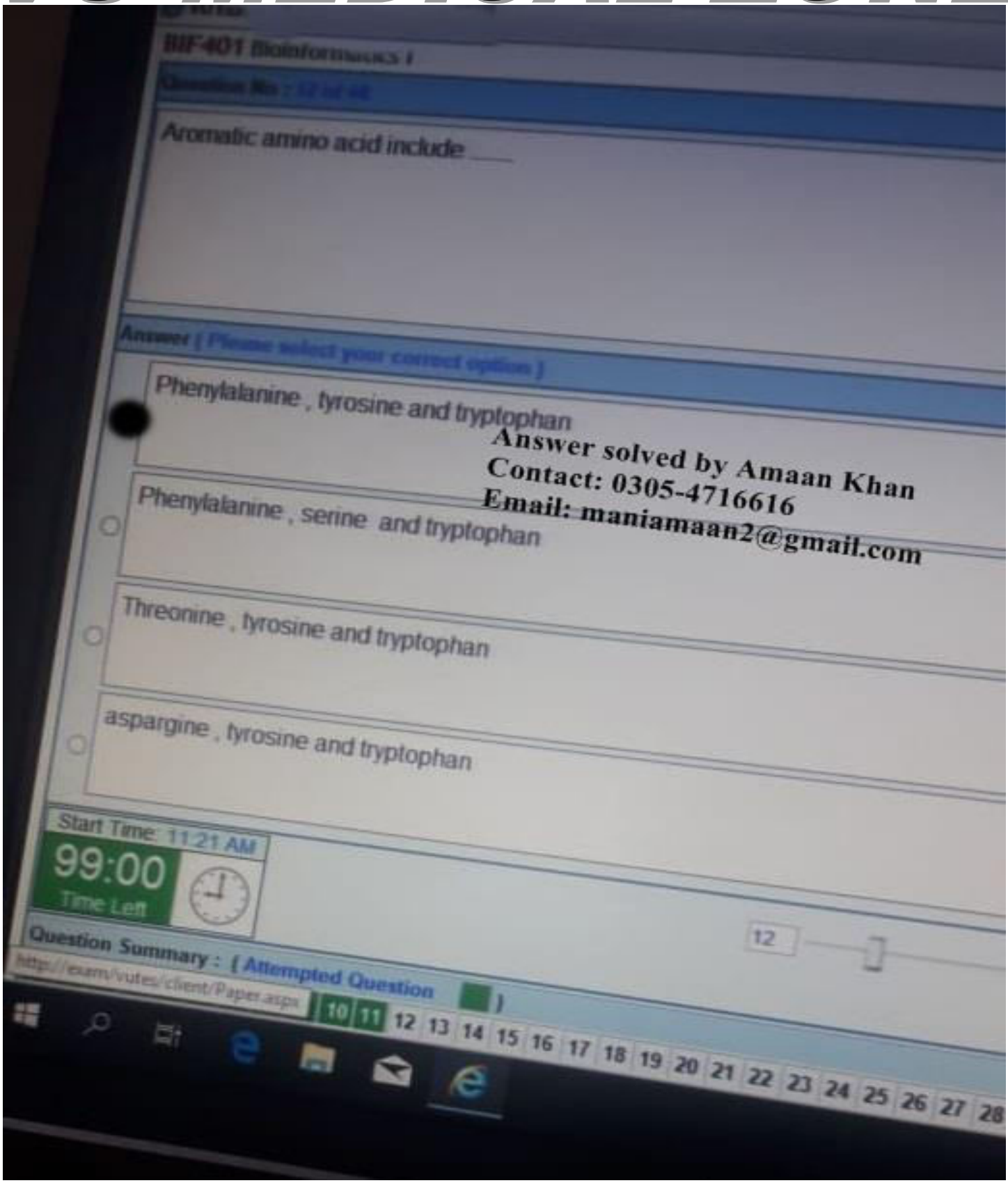
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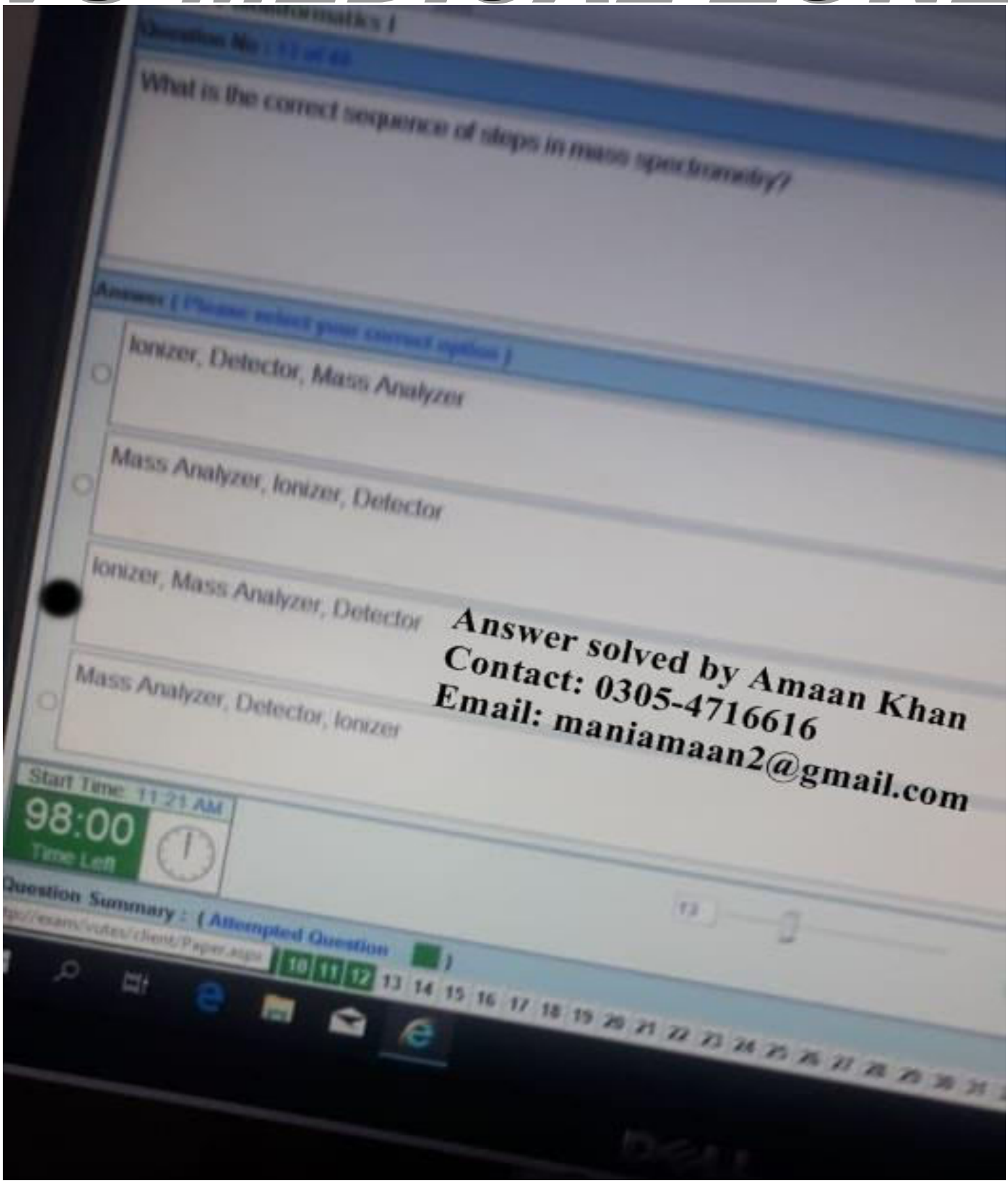
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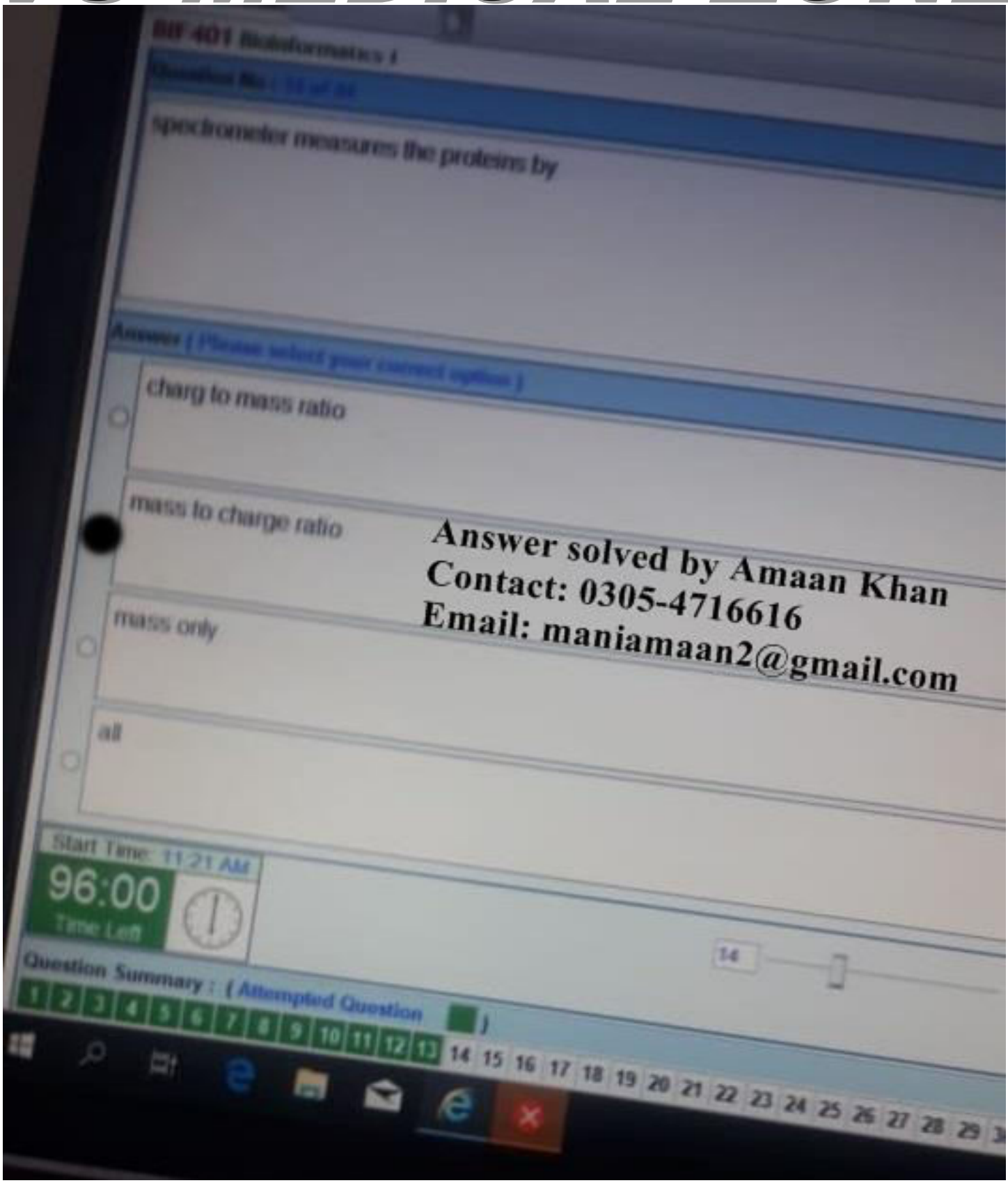
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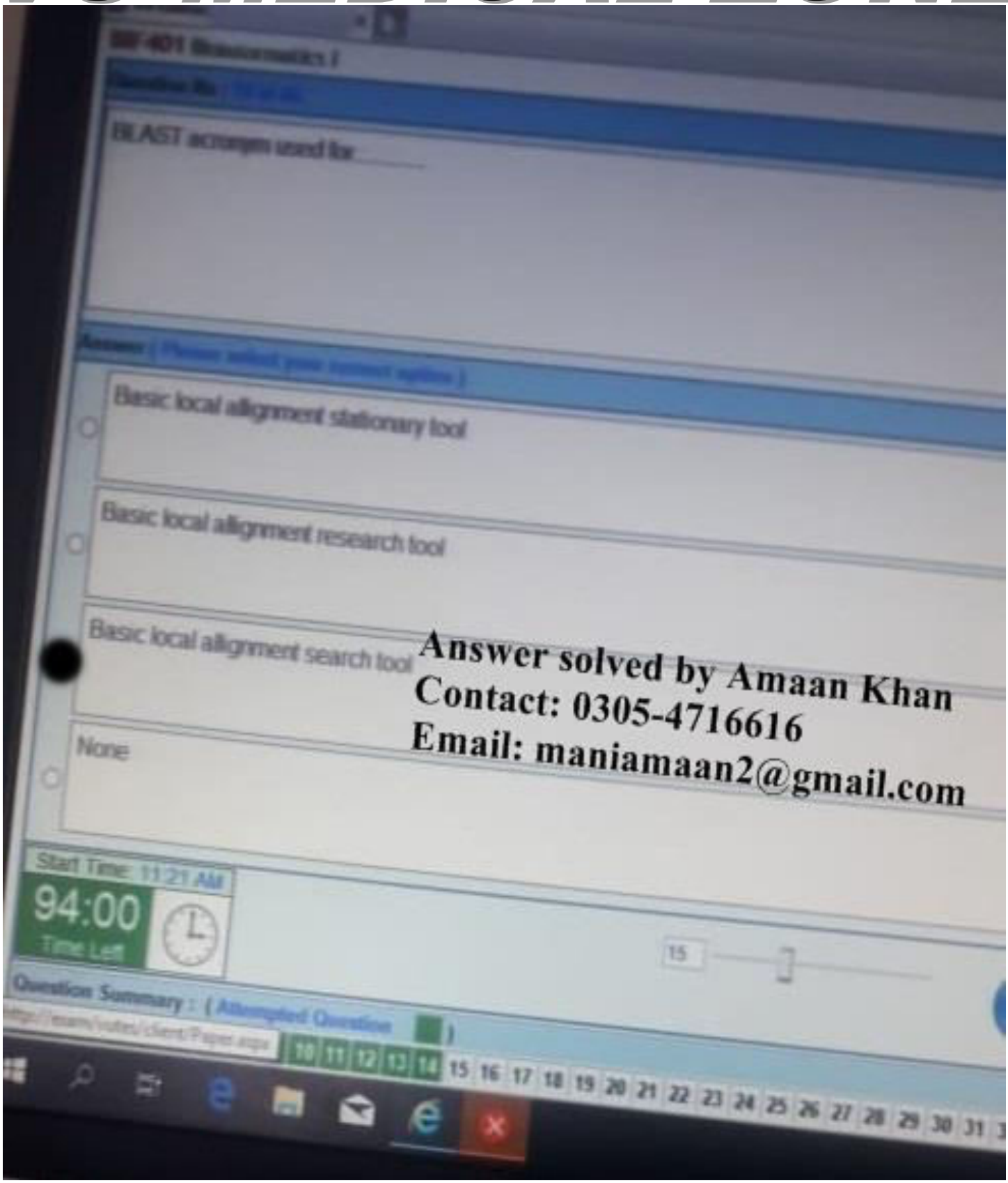
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VU MEDICAL ZONE



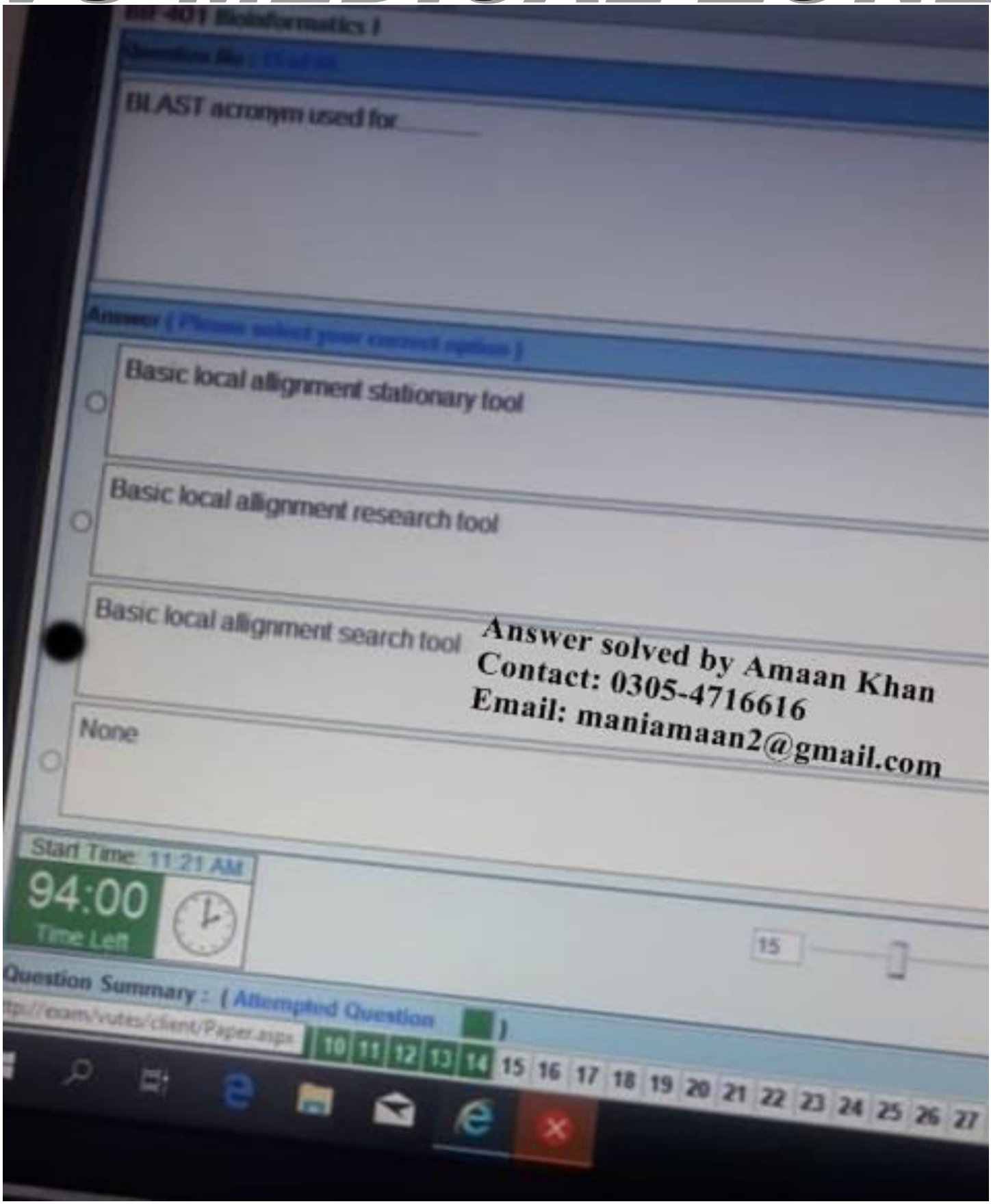
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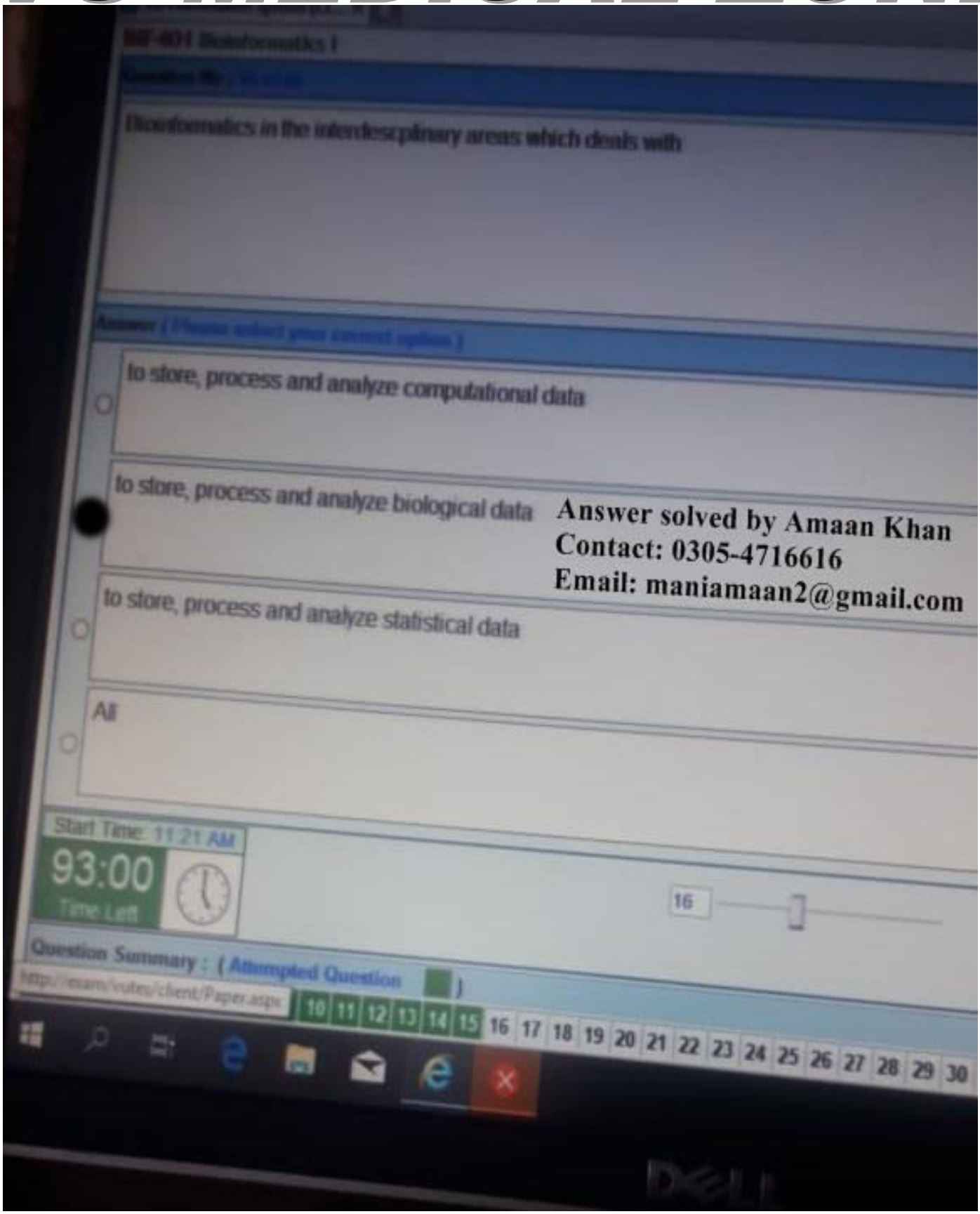
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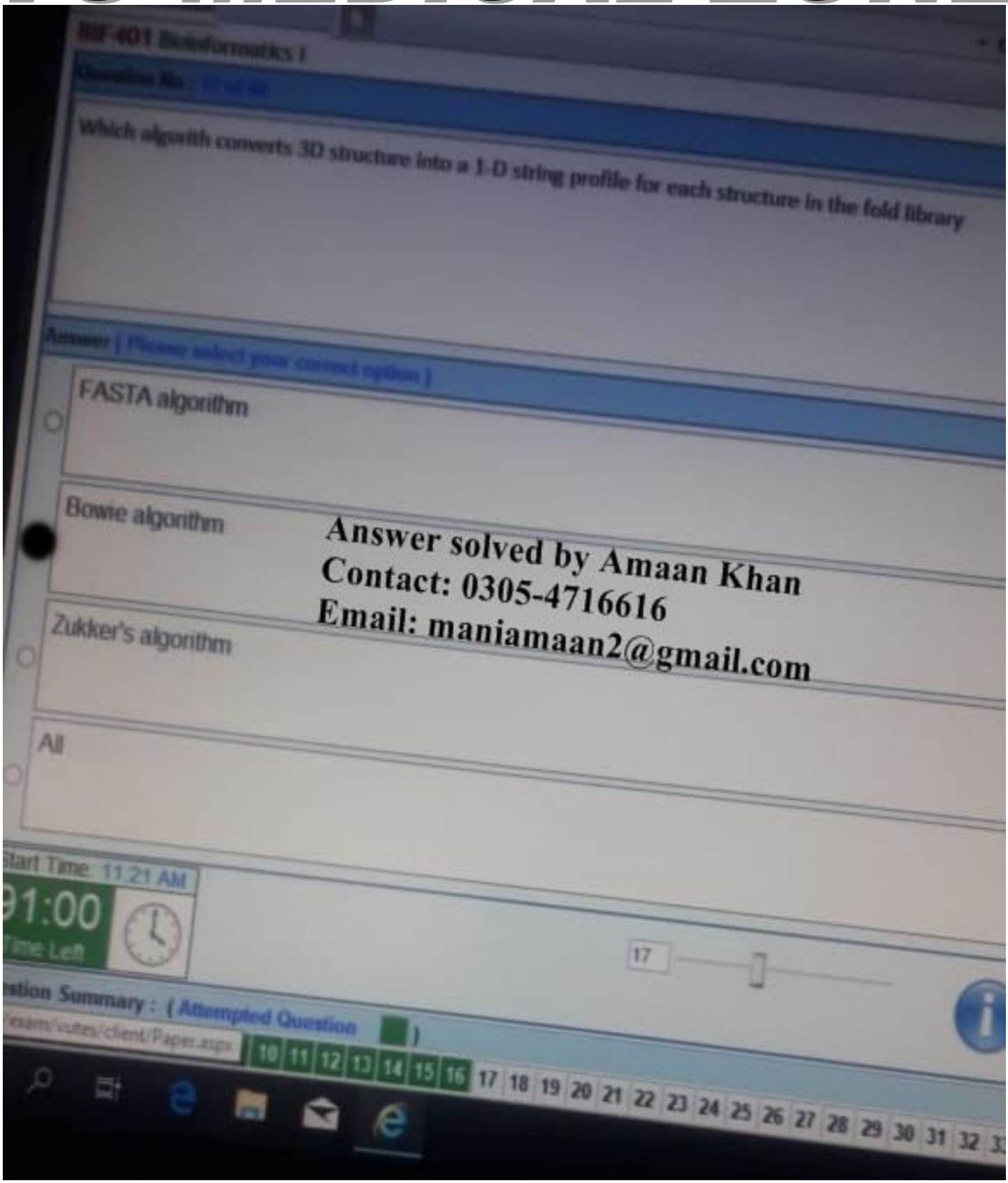
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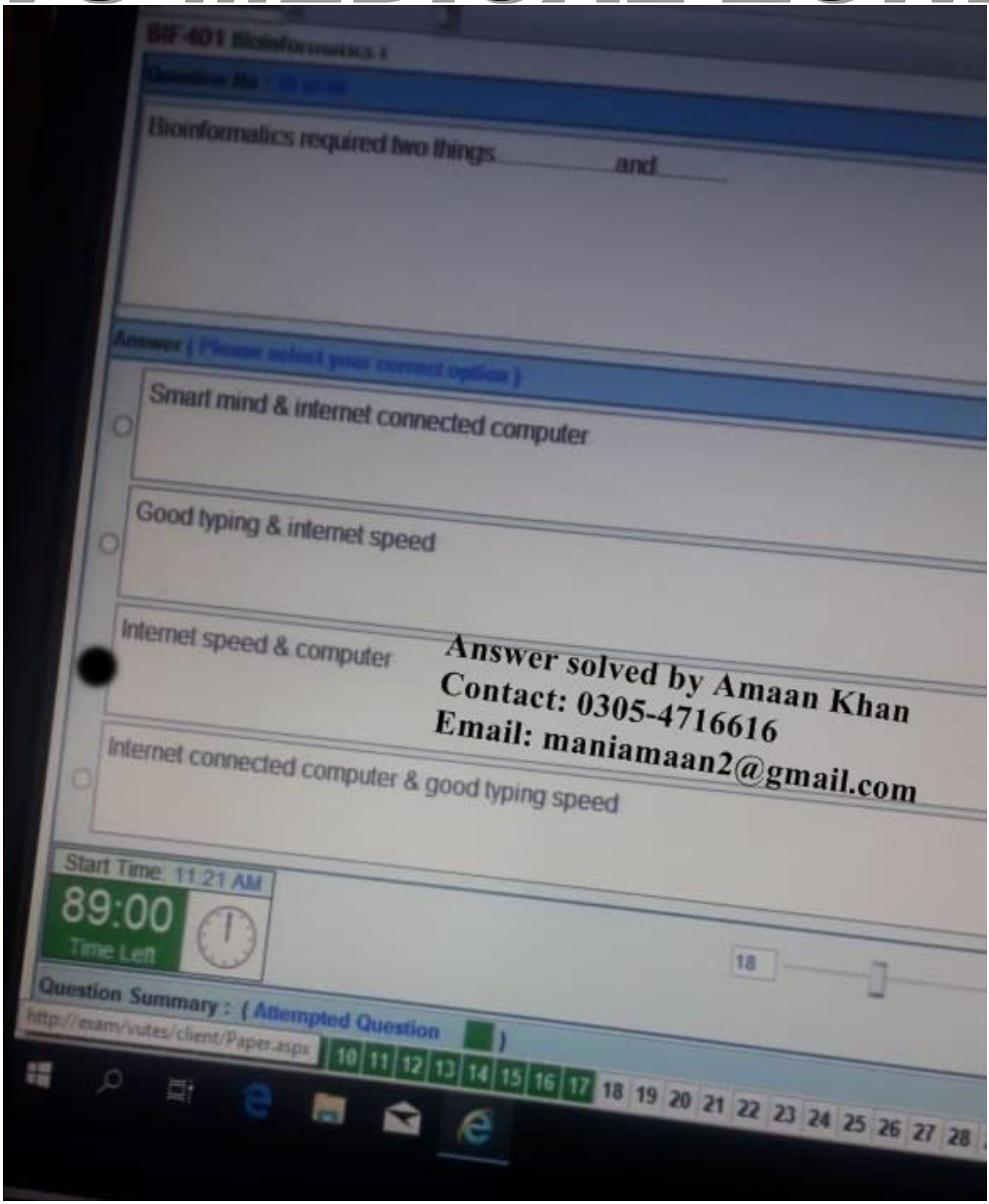
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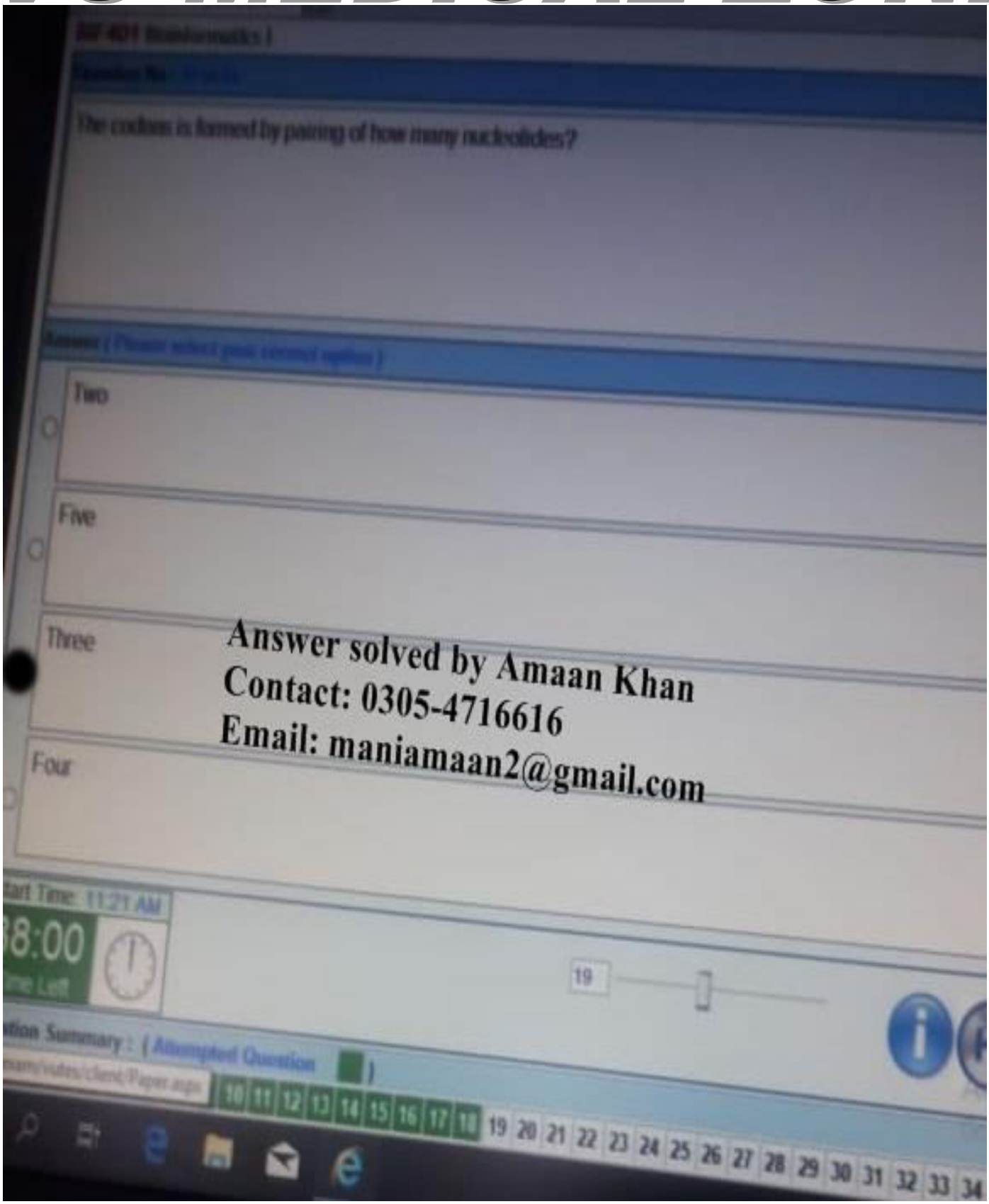
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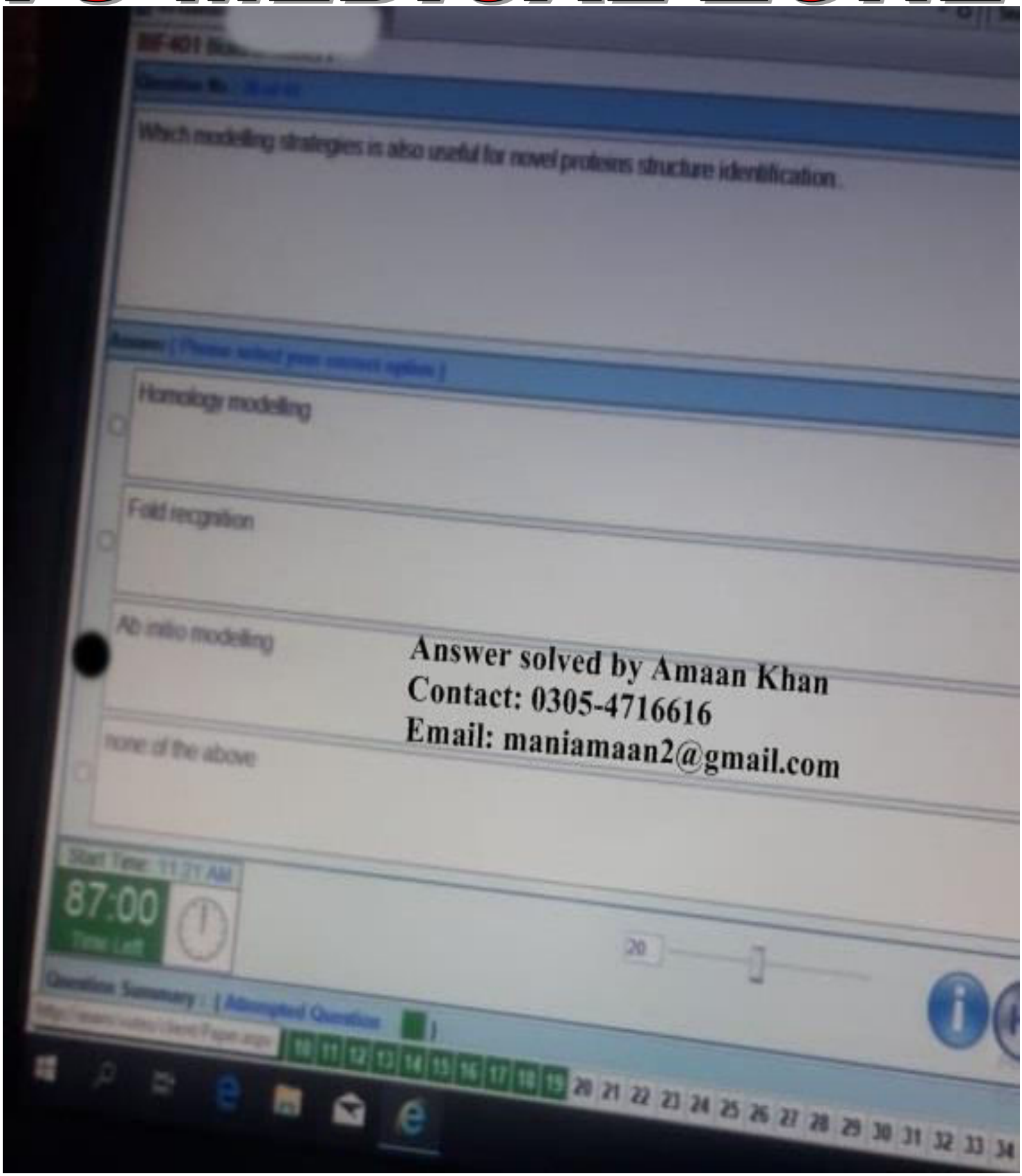
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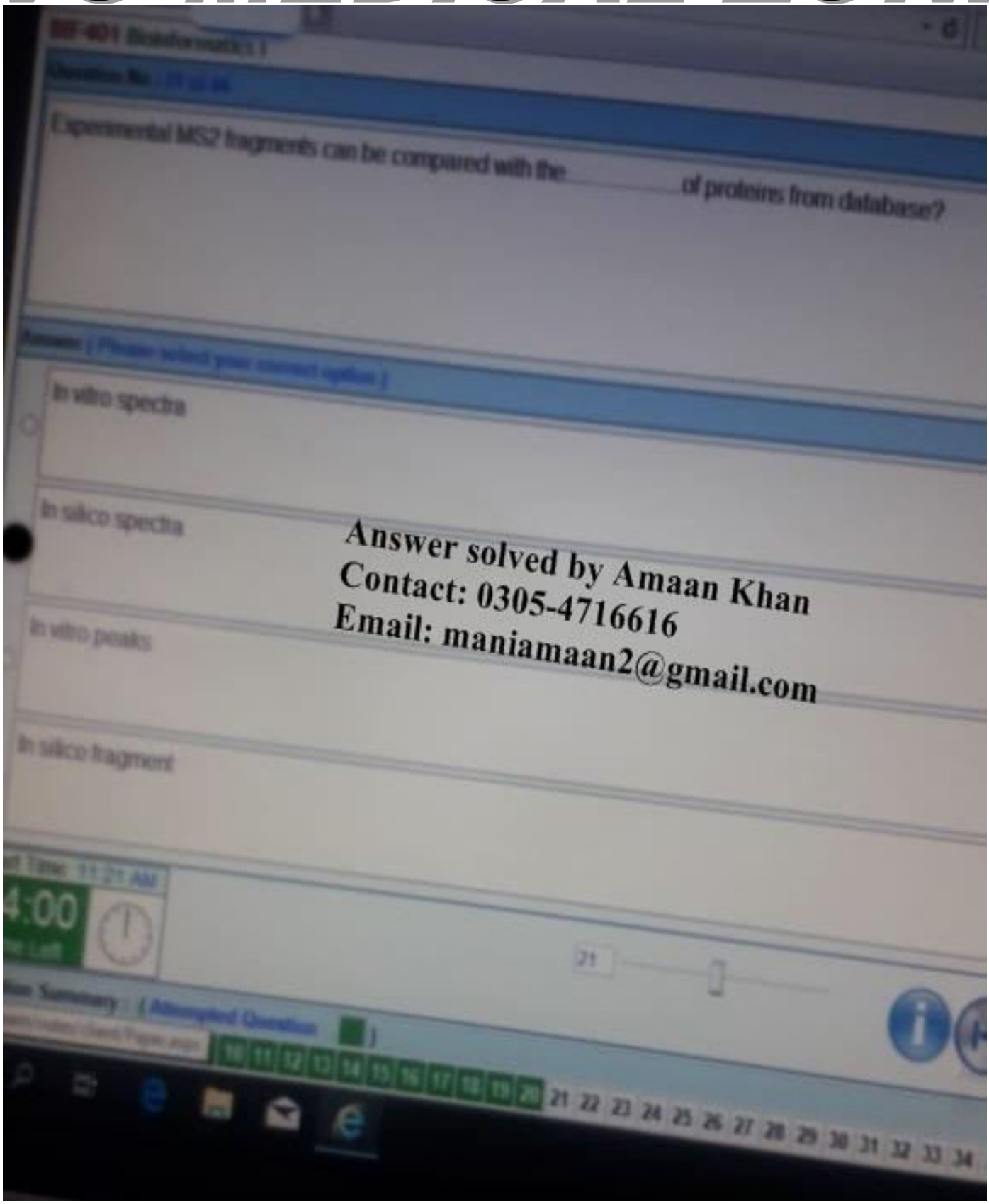
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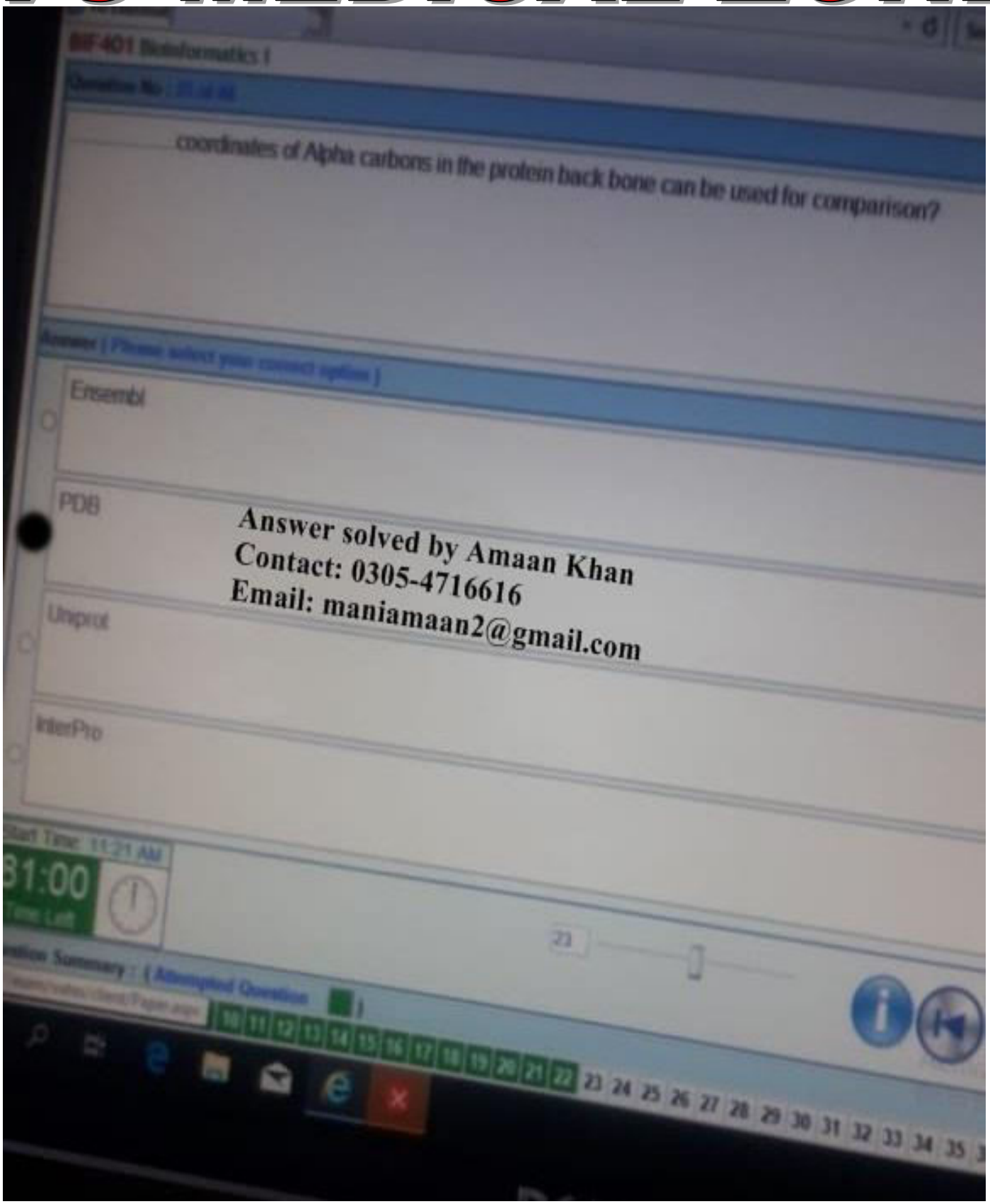
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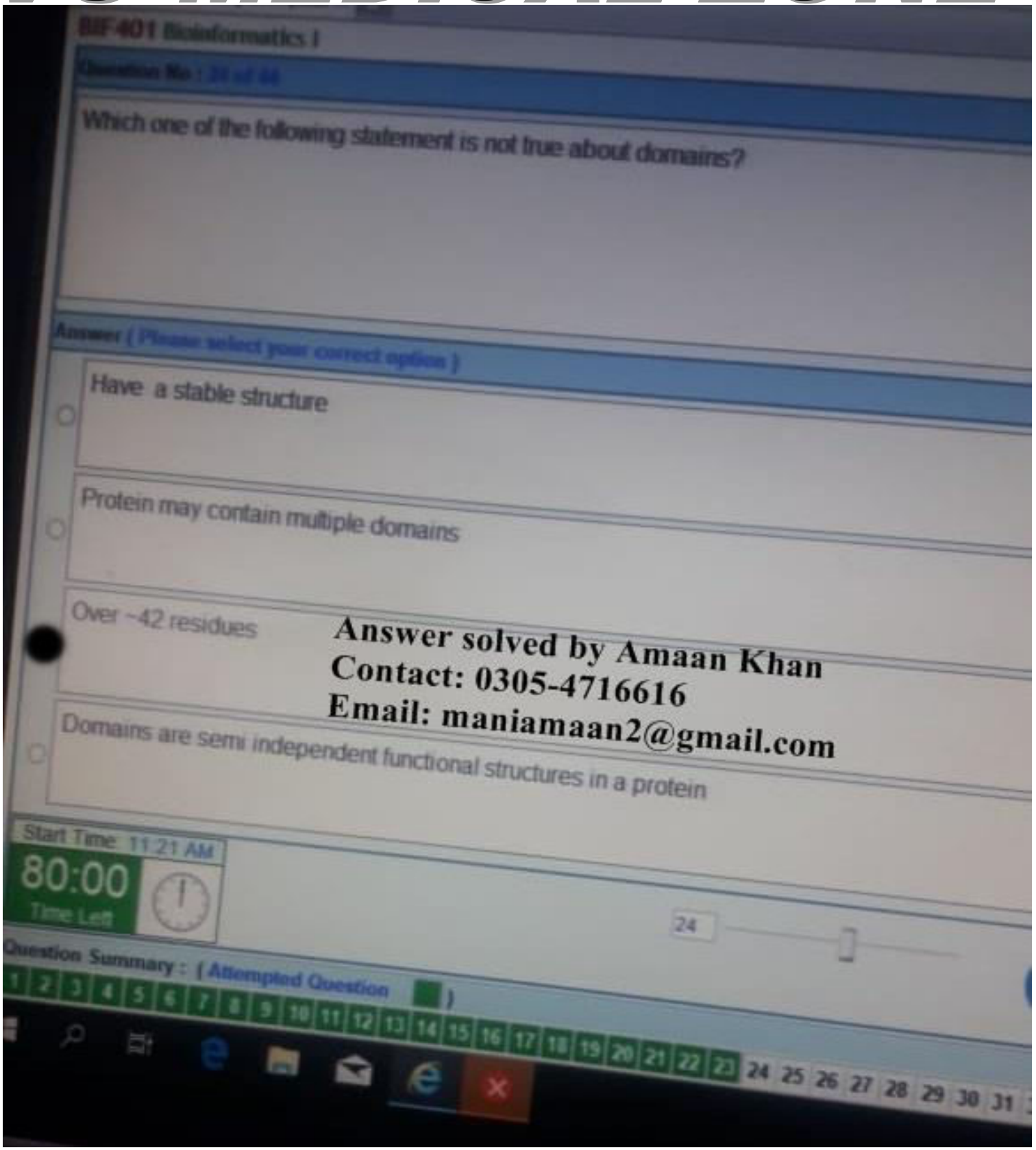
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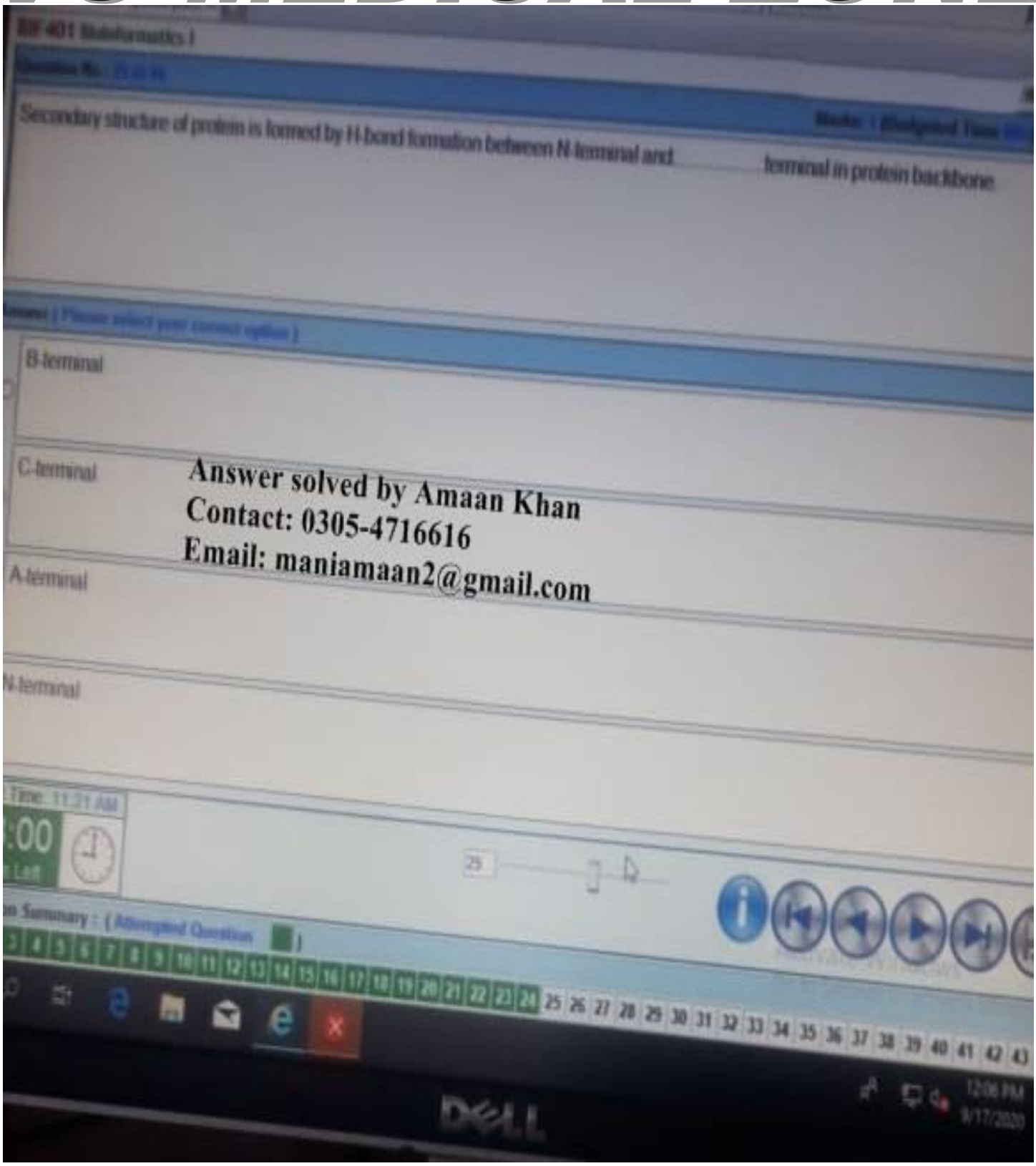
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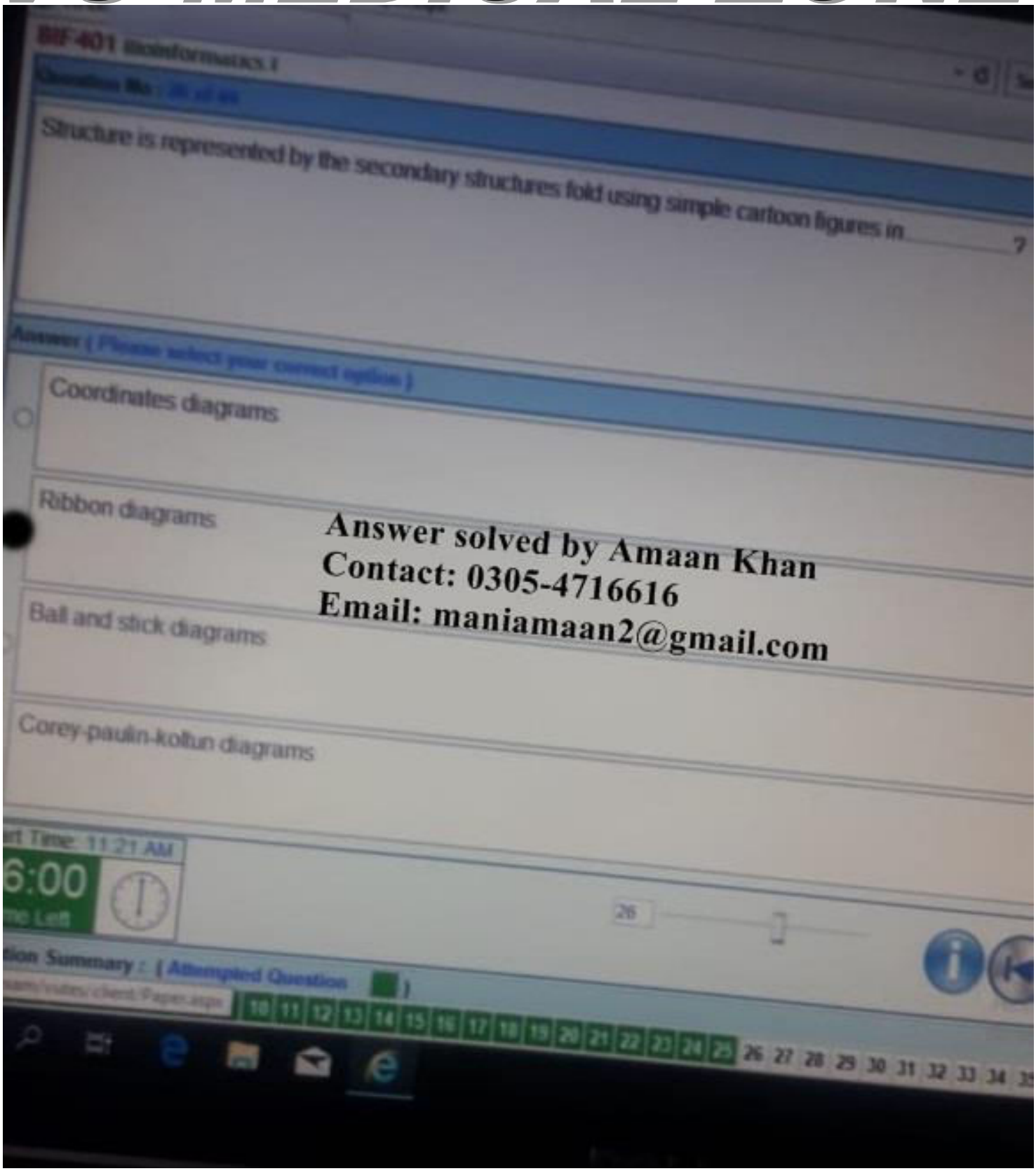
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VU MEDICAL ZONE



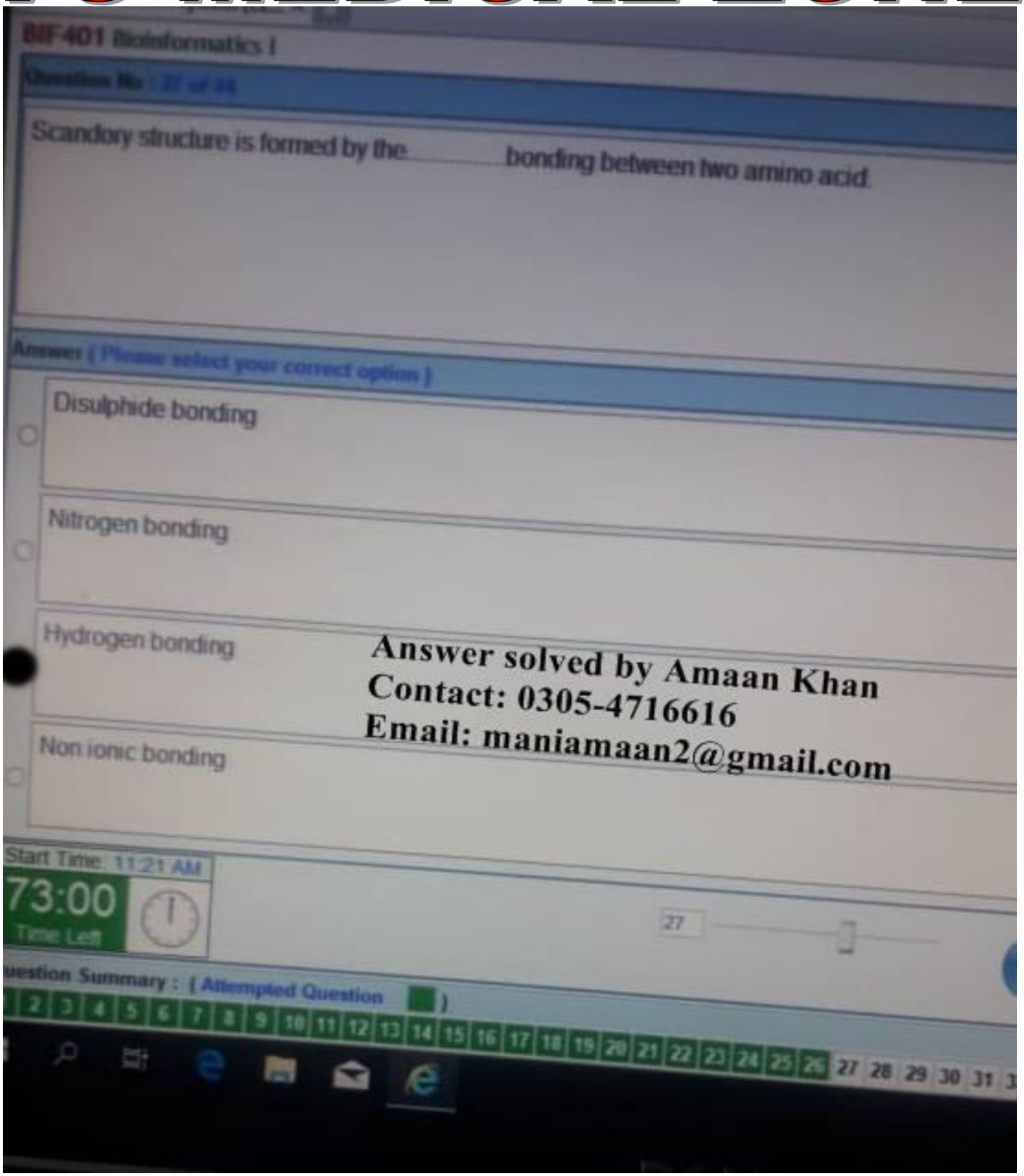
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VU MEDICAL ZONE



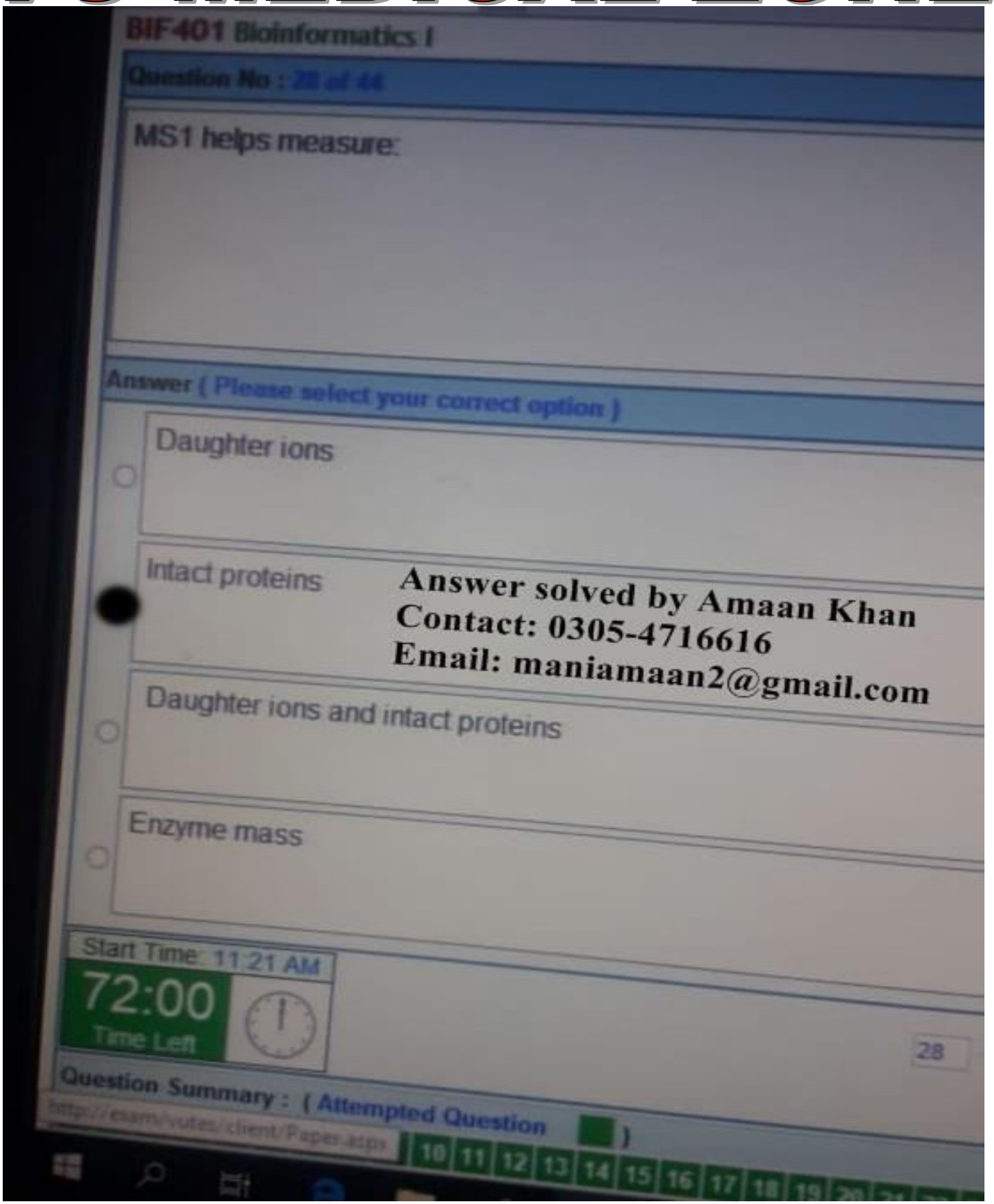
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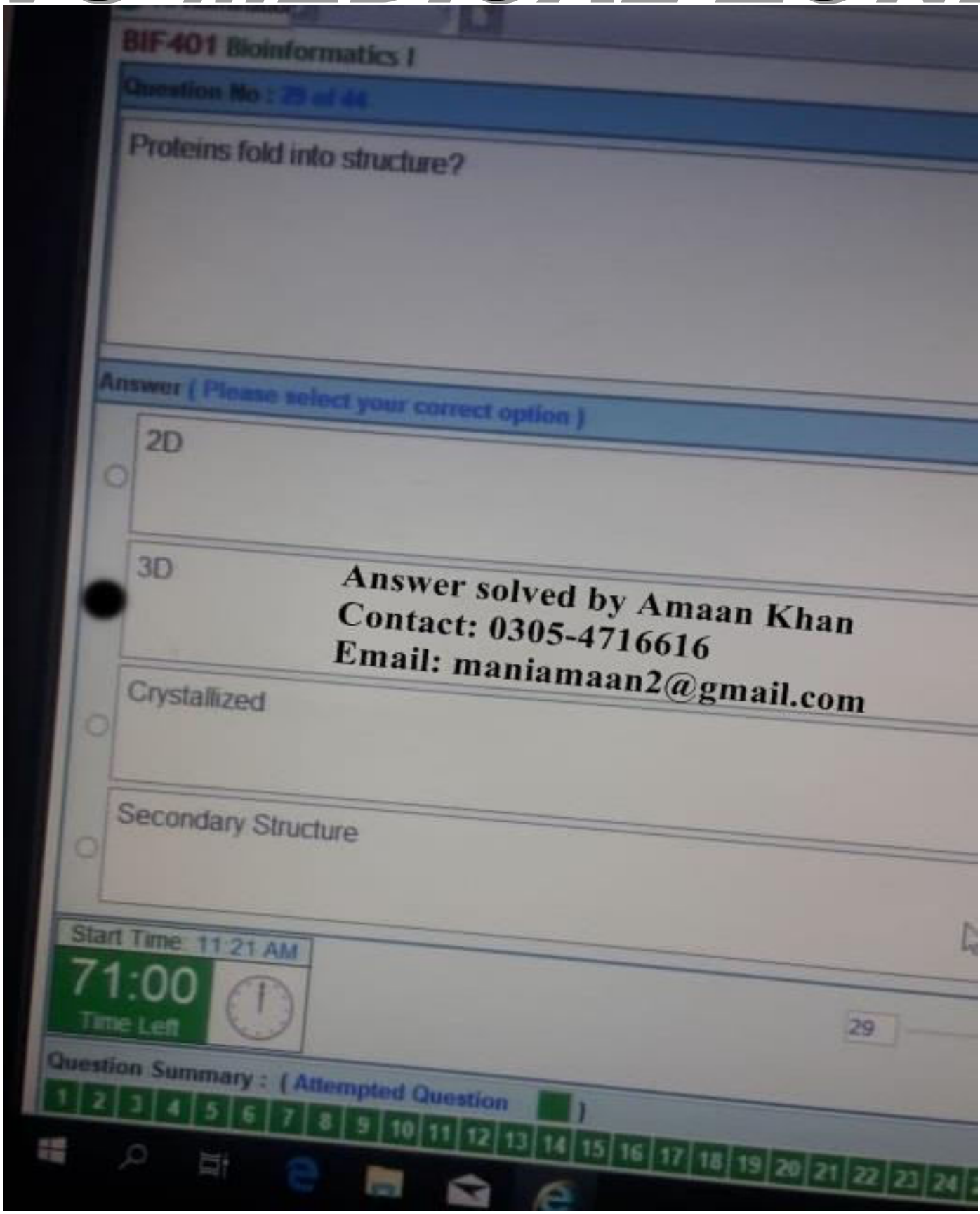
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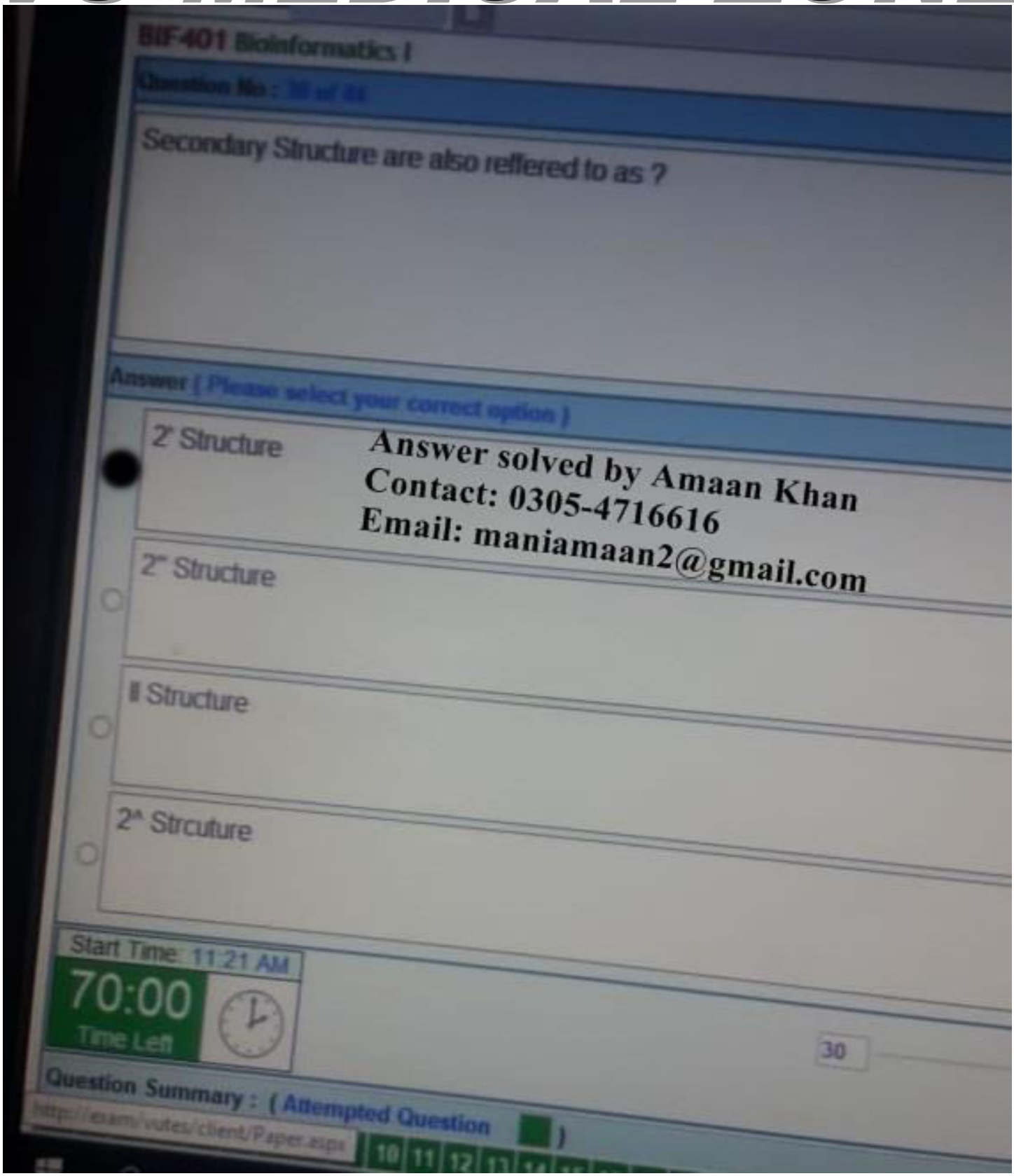
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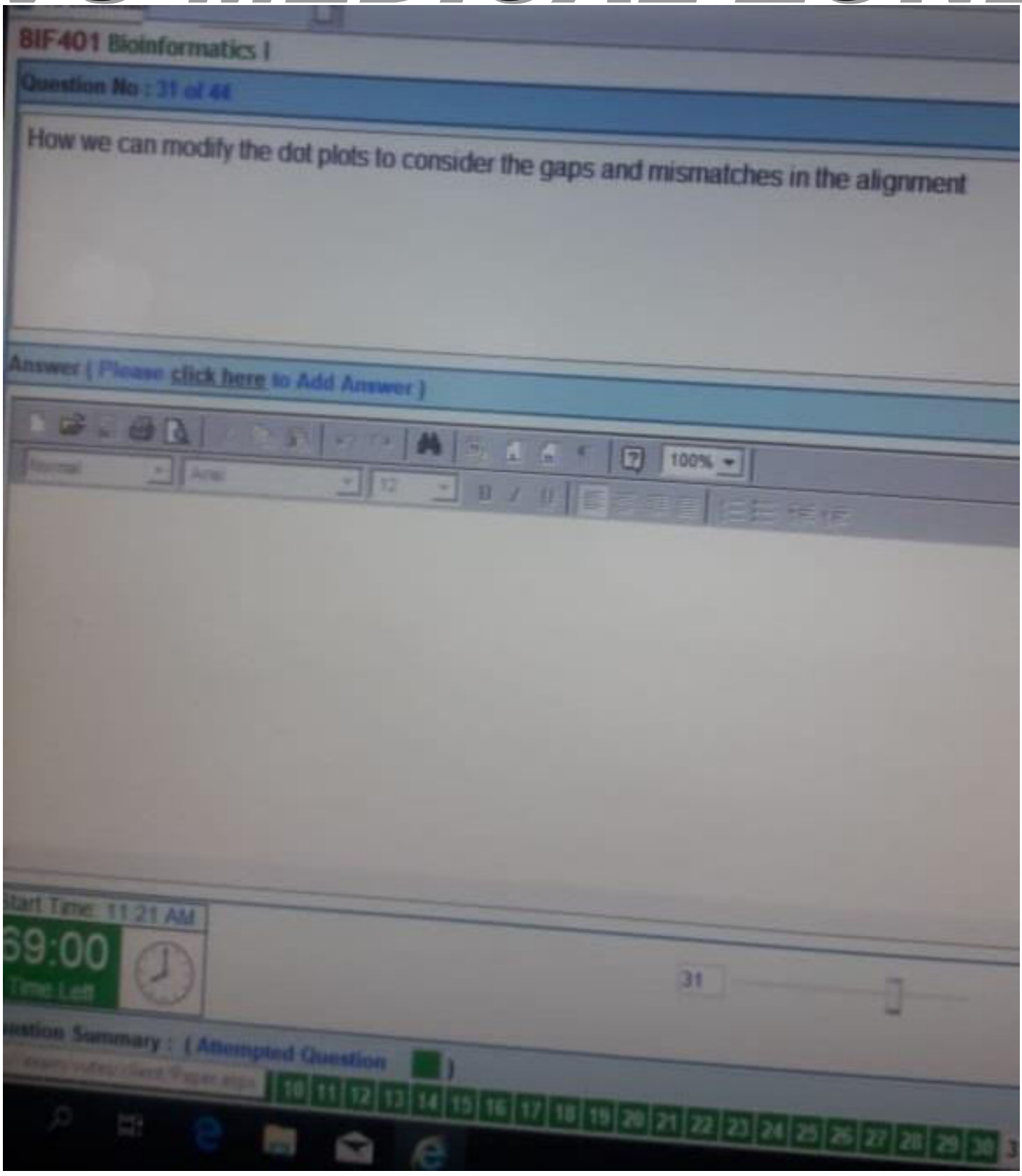
EFFORT BY AMAAN KHAN

VU MEDICAL ZONE



EFFORT BY AMAAN KHAN

VU MEDICAL ZONE

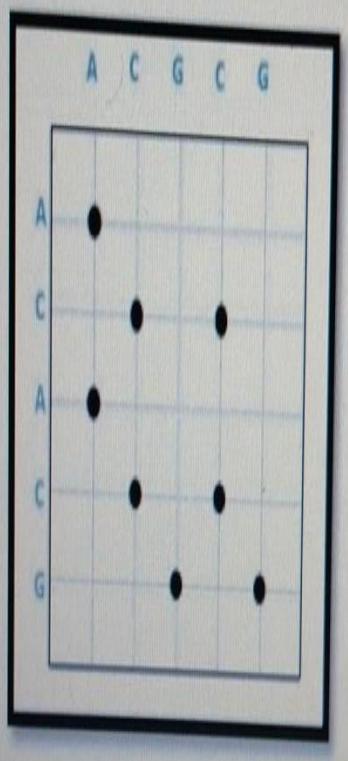


EFFORT BY AMAAN KHAN

VU MEDICAL ZONE

Qno 31

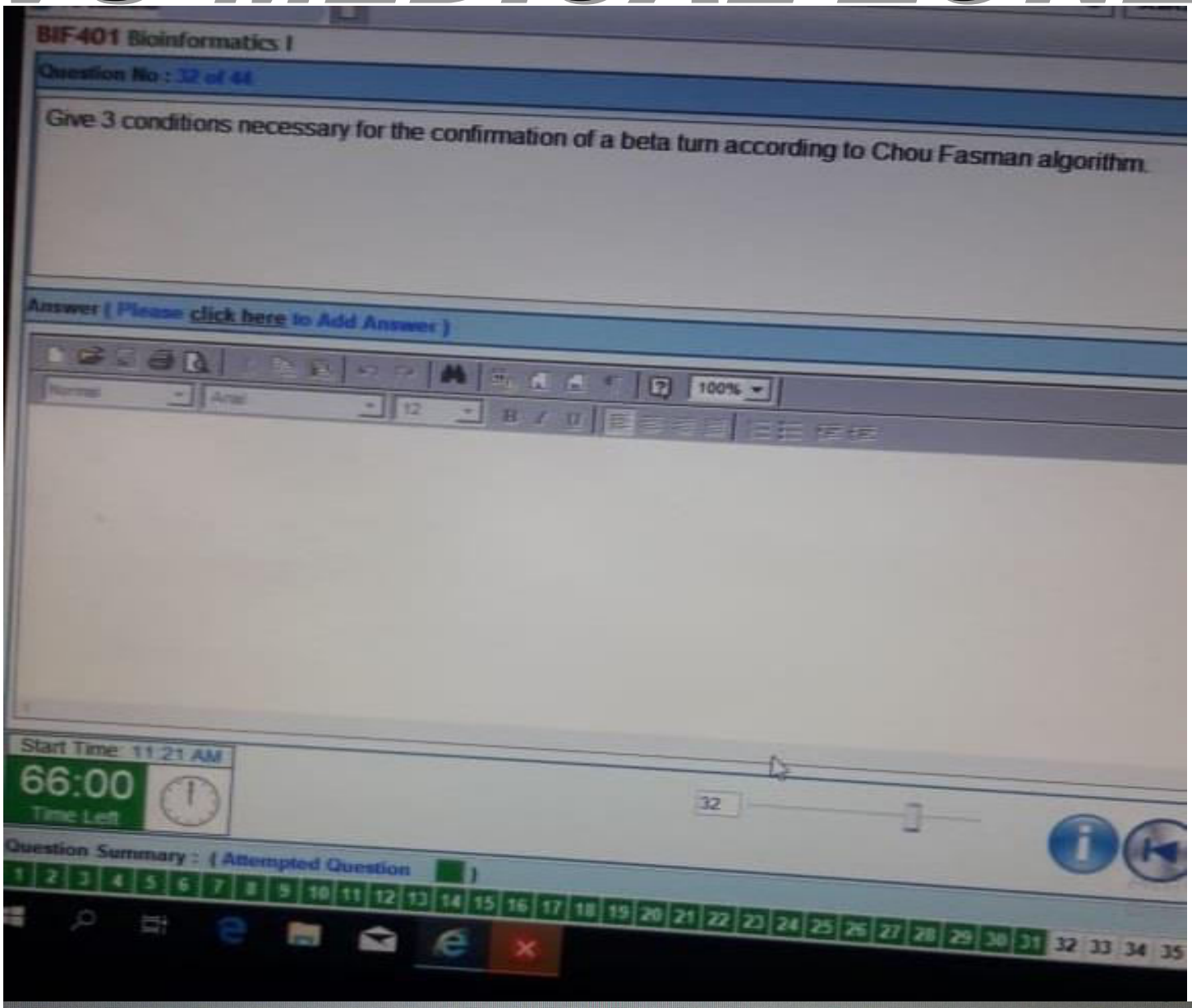
Where one nucleotide or amino acid match with each other the dot is placed in grid position in each row for one time.



Similar dots are match with diagonal pattern and which remain separate differ from similar sequence

EFFORT BY AMAAN KHAN

VU MEDICAL ZONE



We can evaluate such regions for Beta Sheets. Let us see step by stop how to find a beta sh
and how to differentiate them from alpha helices.

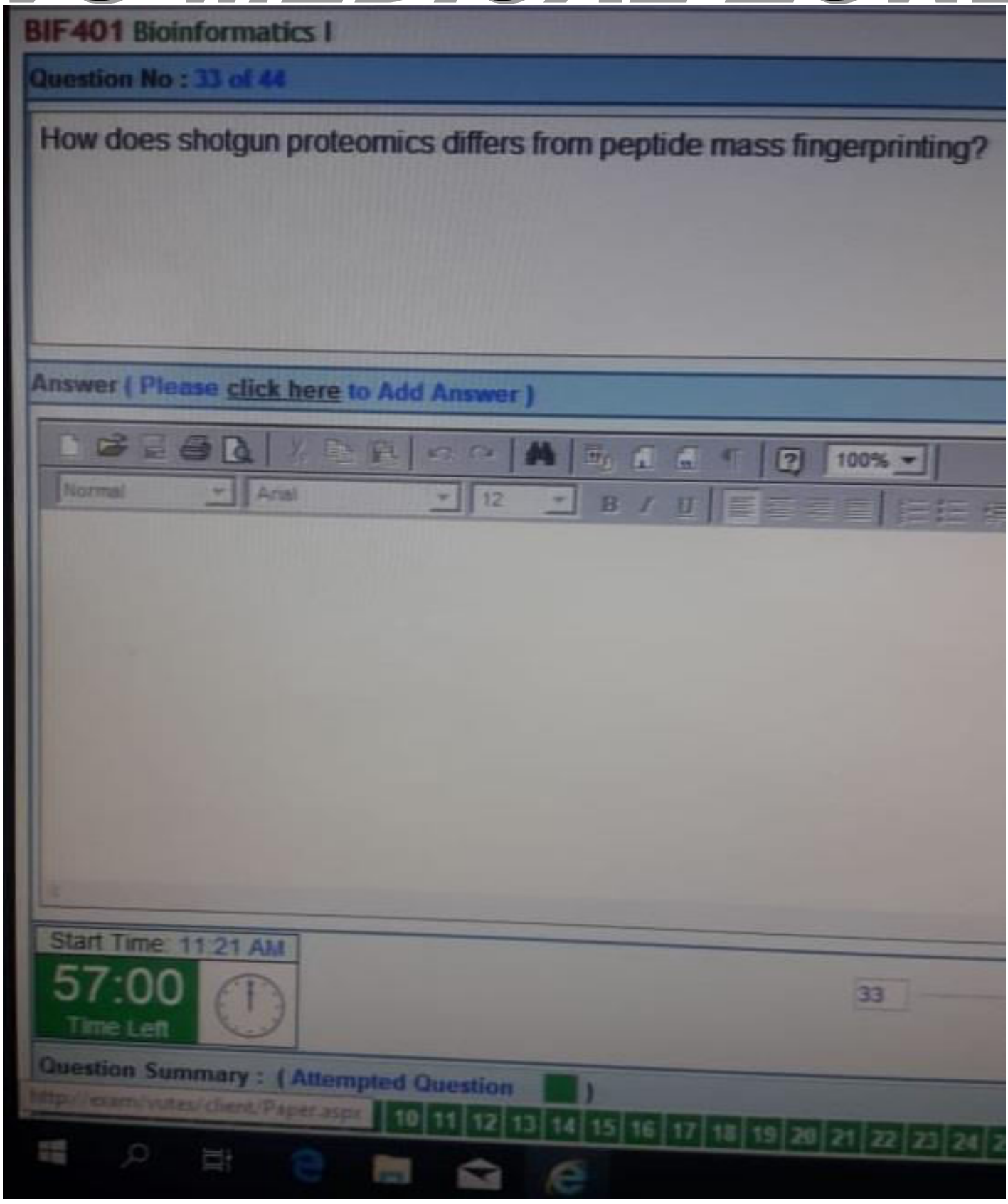
Qno 32

Scan the sequence to identify regions where:

- 3 out of 5 amino acids have $P(\beta) > 1.0$
- That region is declared as beta sheet
- Extend beta sheet to both sides until
  4 contiguous residues average $P(\beta) < 1.0$
- That is declared end of the beta sheet
- Those regions are finalized as beta-sheets which have average $P(\beta) > 1.05$ and the average $P(\beta) > P(\alpha)$ for that region.

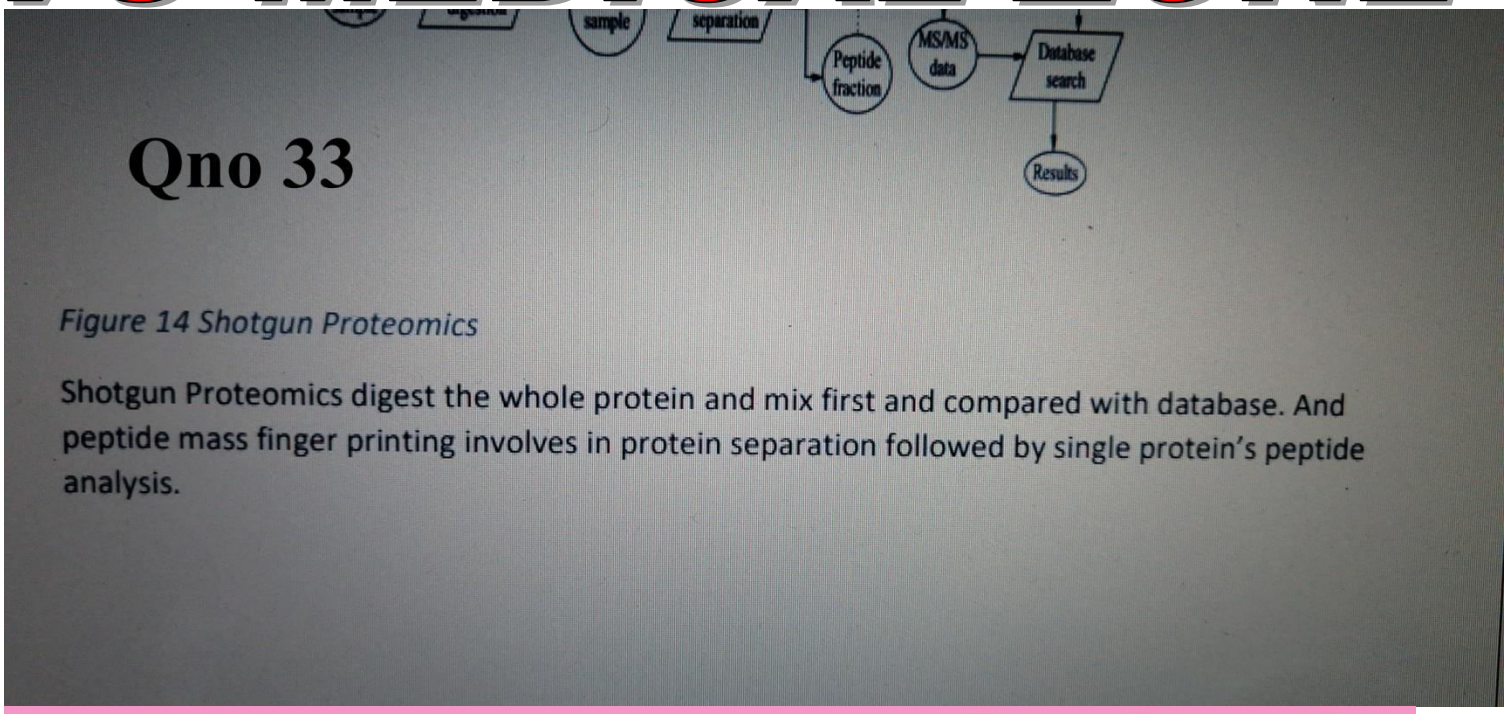
EFFORT BY AMAAN KHAN

VU MEDICAL ZONE



EFFORT BY AMAAN KHAN

VU MEDICAL ZONE



Qno 33

Figure 14 Shotgun Proteomics

Shotgun Proteomics digest the whole protein and mix first and compared with database. And peptide mass finger printing involves in protein separation followed by single protein's peptide analysis.

- Denaturation is associated with increase in ionizable and sulfhydryl groups of protein. This is due to loss of hydrogen and disulfide bonds.

Qno 44

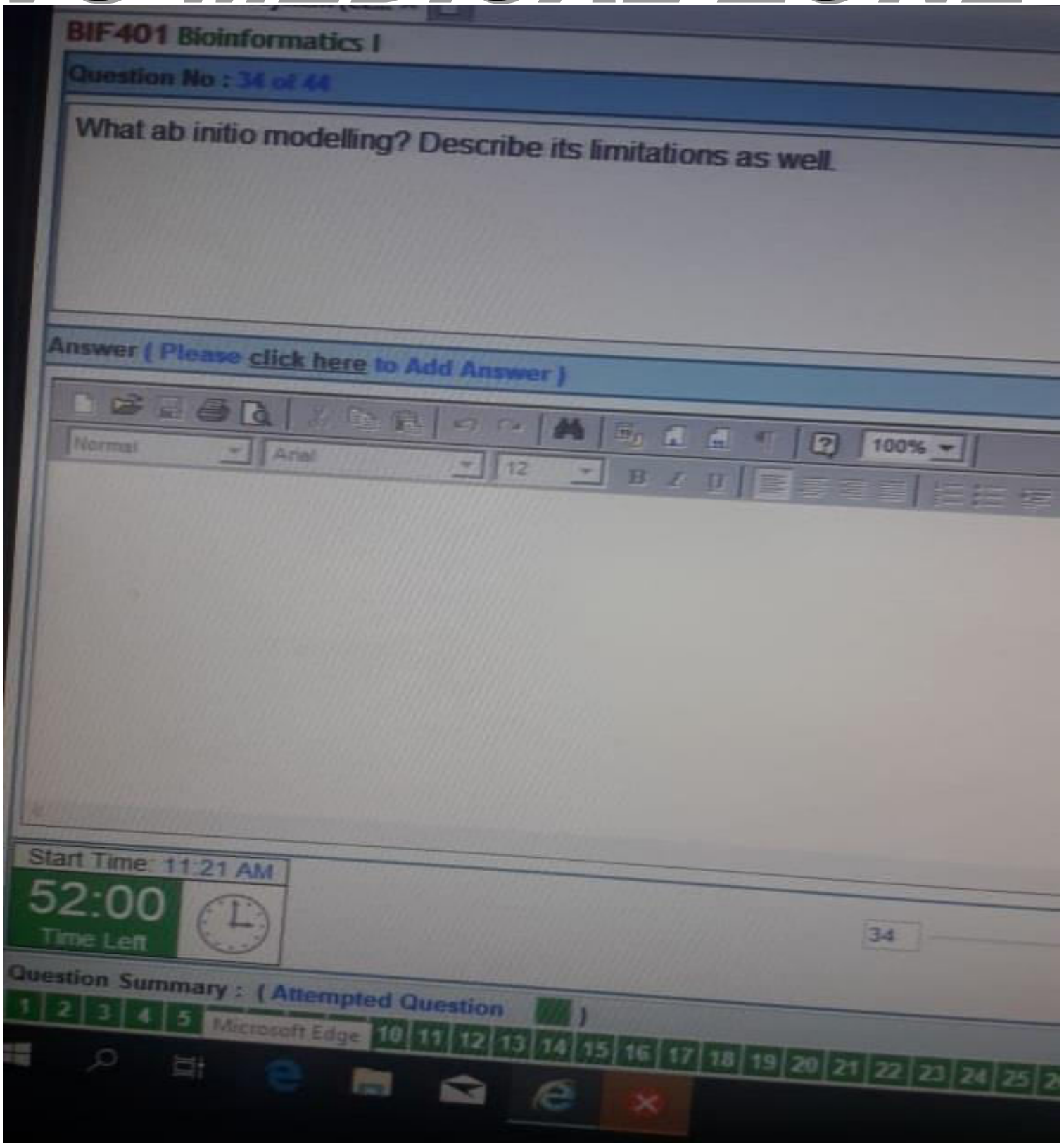
- Denatured protein is more easily digested. This is due to increased exposure of peptide bonds to enzymes. Cooking causes protein denaturation and therefore, cooked food (protein) is more easily digested.

Advantages of protein denaturation:

- The study of denaturation of proteins helps in the field of proteomics.
- To determine the concentration of protein in any given food samples.

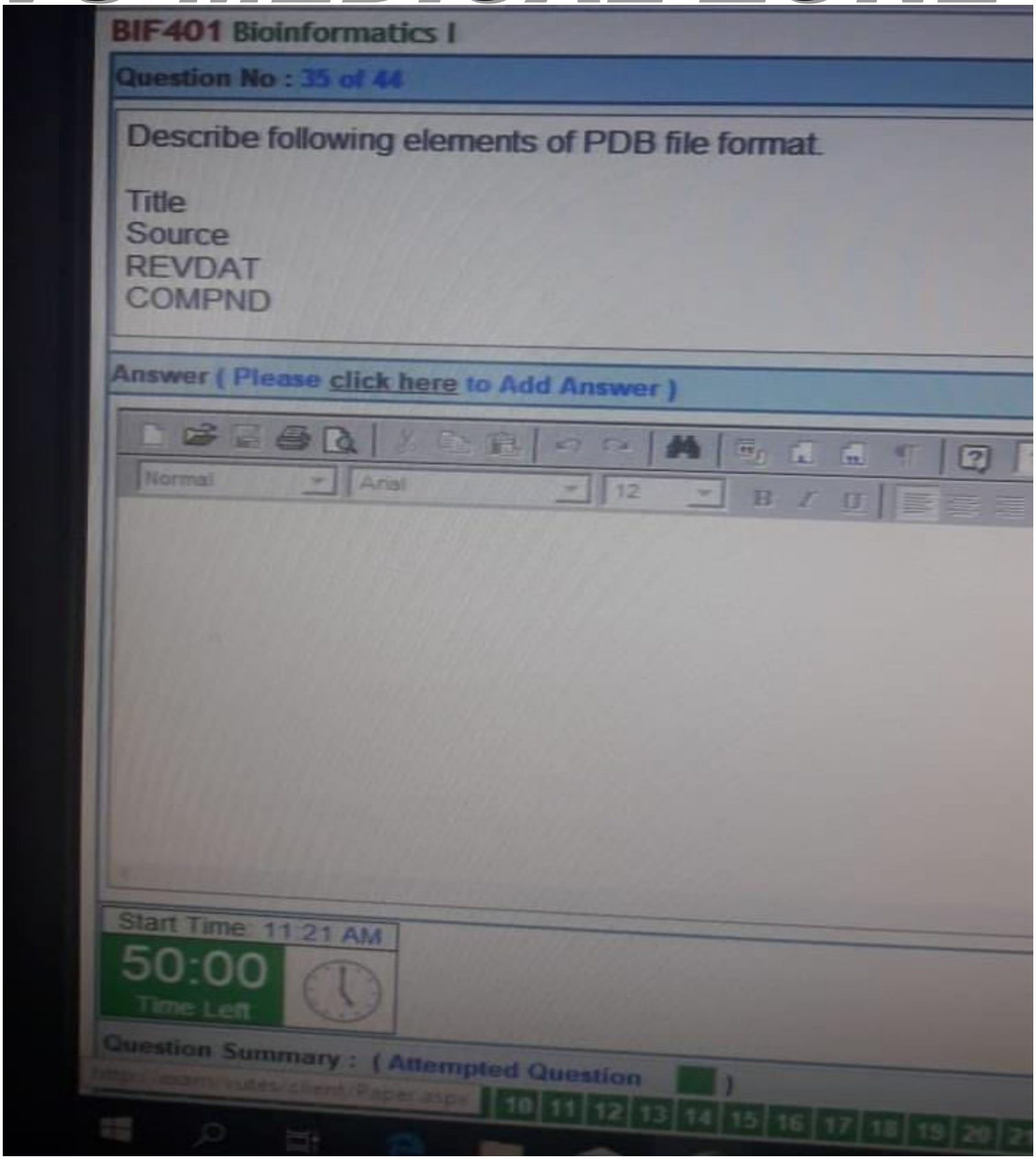
EFFORT BY AMAAN KHAN

VU MEDICAL ZONE



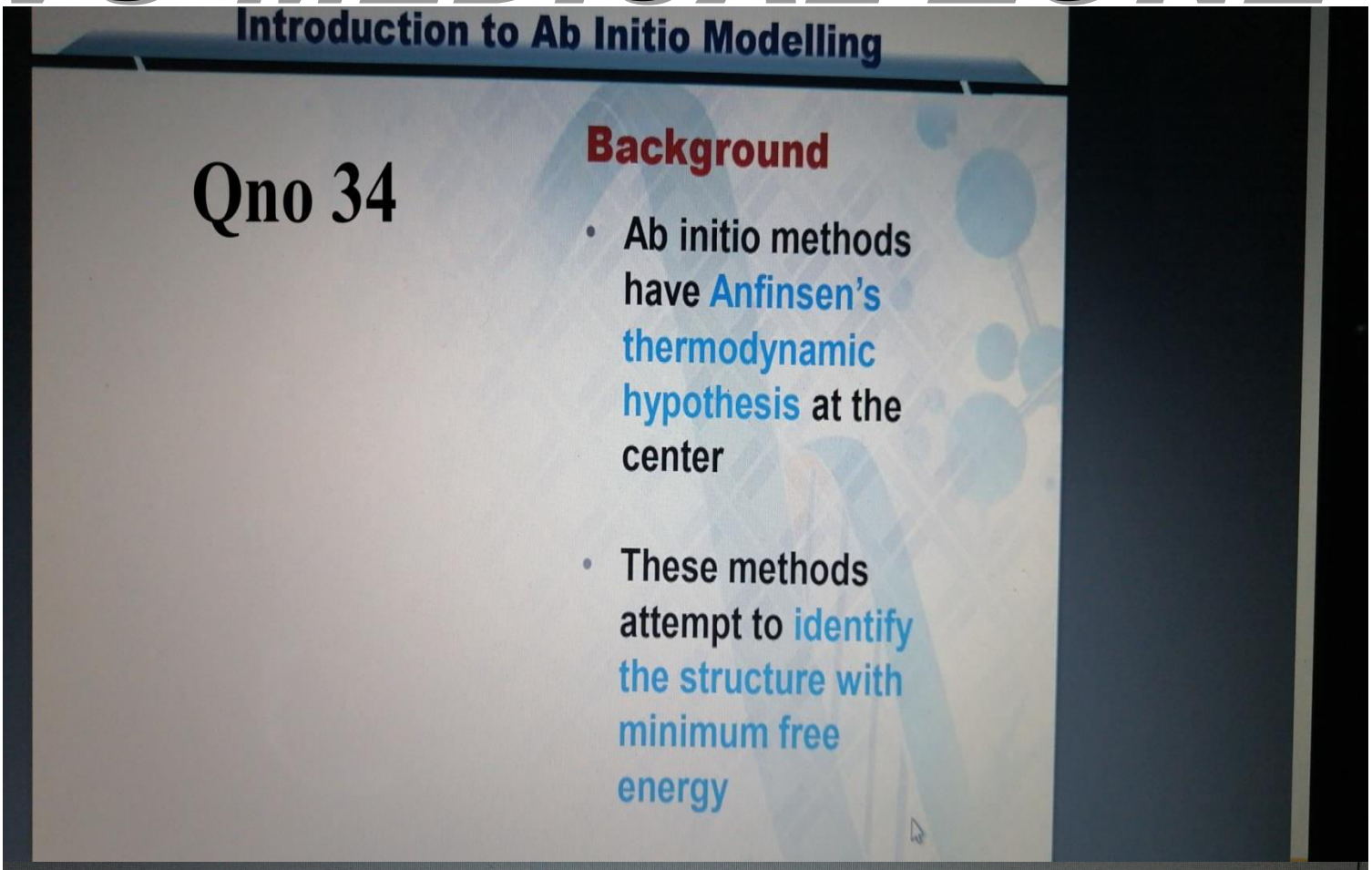
EFFORT BY AMAAN KHAN

VU MEDICAL ZONE



EFFORT BY AMAAN KHAN

VU MEDICAL ZONE



Qno 35

HEADER - Contains a brief description of the structure, the date and the PDB ID code.

TITLE - The title of the structure.

COMPND - Brief details of the structure.

SOURCE - Identifies which organism the structure came from.

KEYWDS - Lists a set of useful words/phrases that describe the structure.

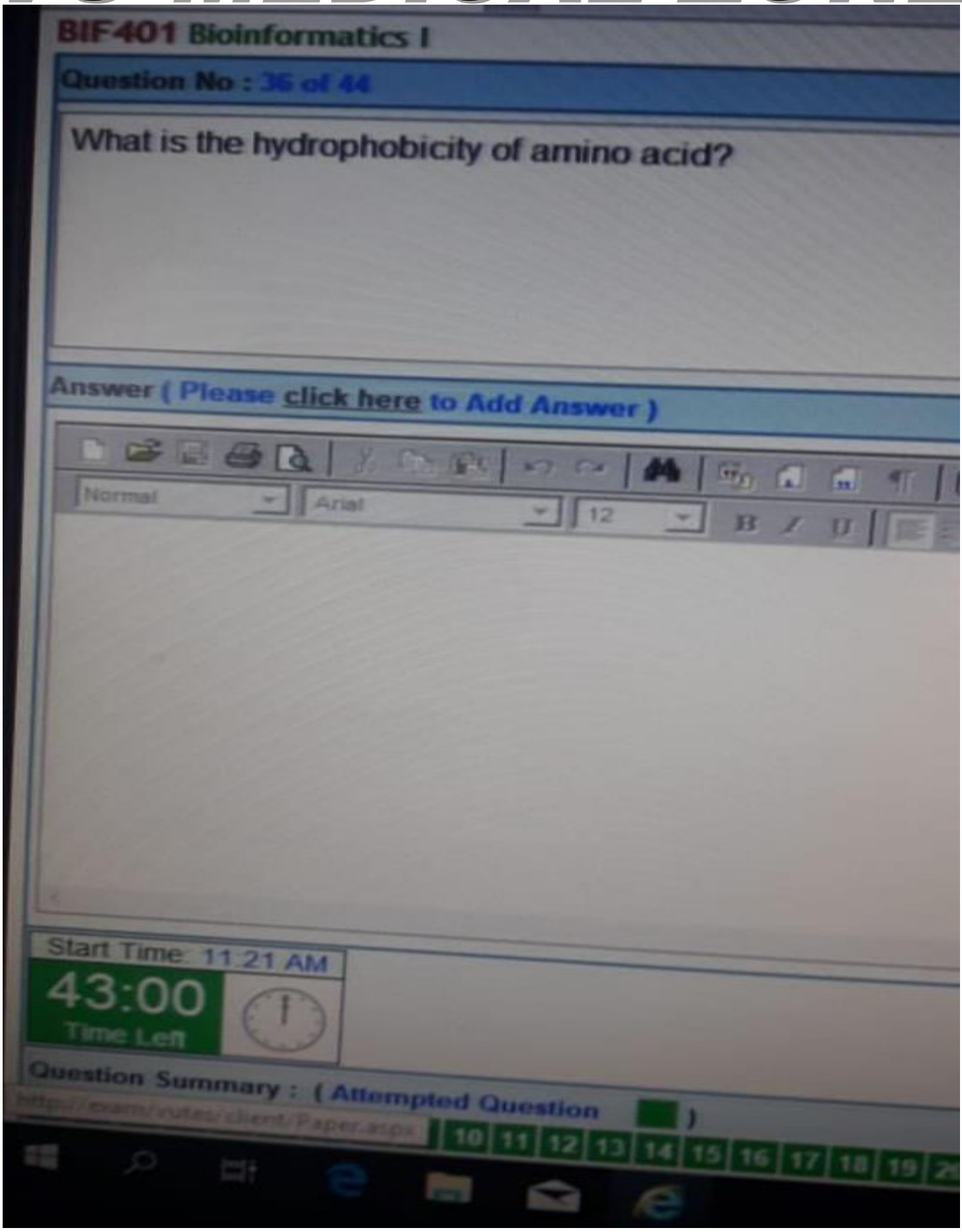
AUTHOR - The scientists depositing the structure.

REVDAT - The date of the last revision.

Figure 49 PDB File Format

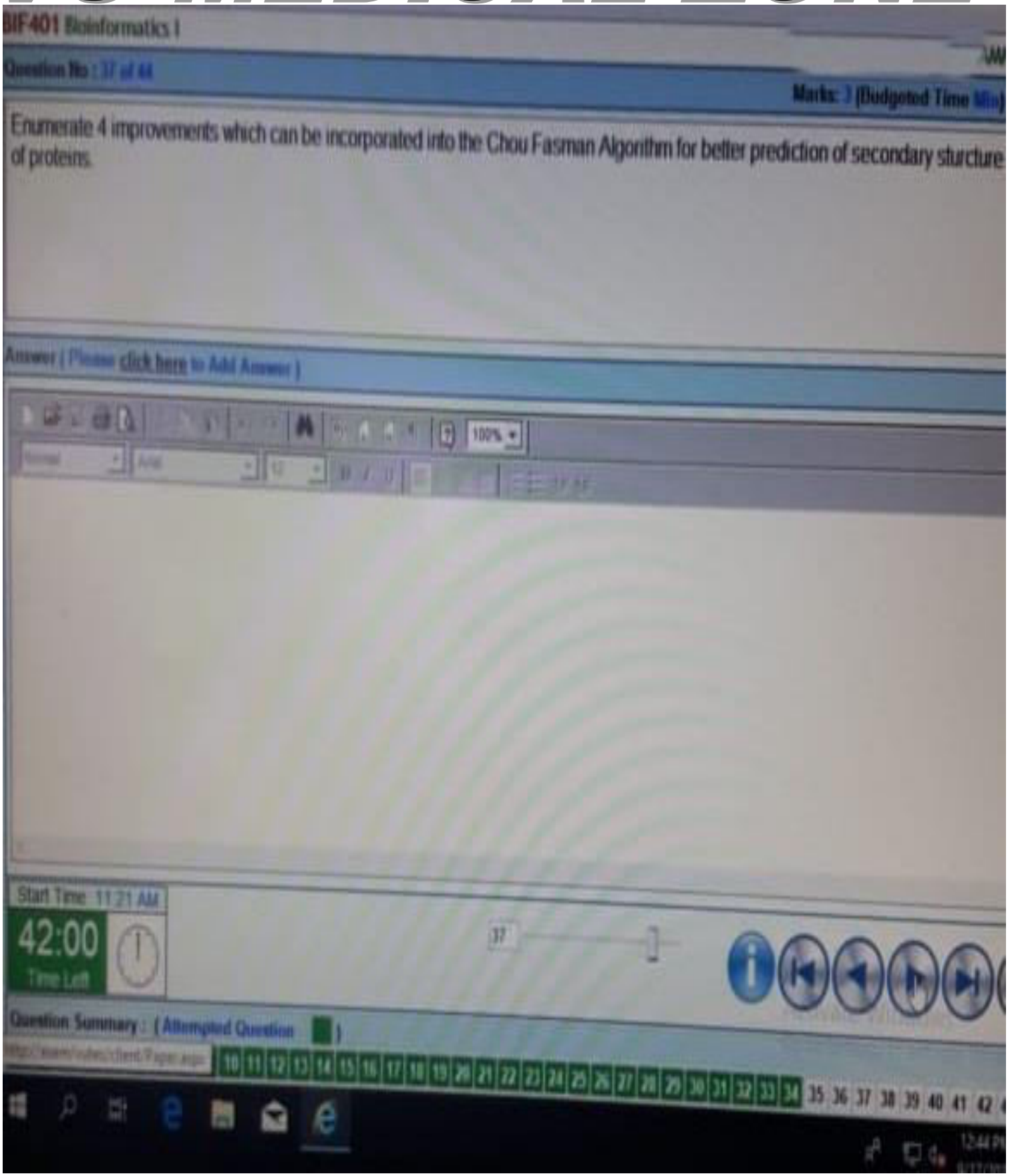
EFFORT BY AMAAN KHAN

VU MEDICAL ZONE



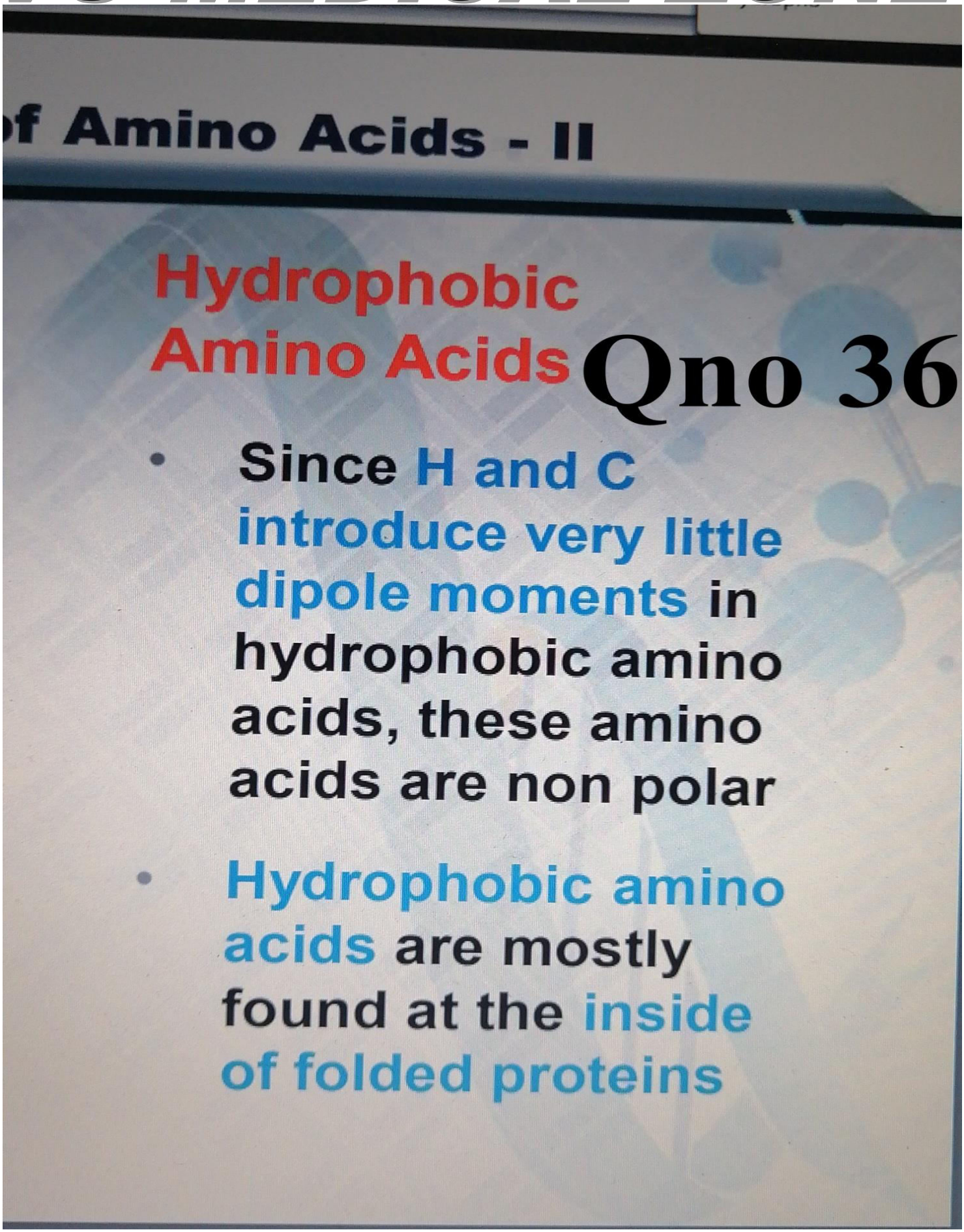
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VU MEDICAL ZONE



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VU MEDICAL ZONE



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VU MEDICAL ZONE

recalculated with the latest protein data sets.

IMPROVEMENTS

Qno 37

Special consideration for:

- Nucleation regions
- Membrane proteins
- Hydrophobic domains

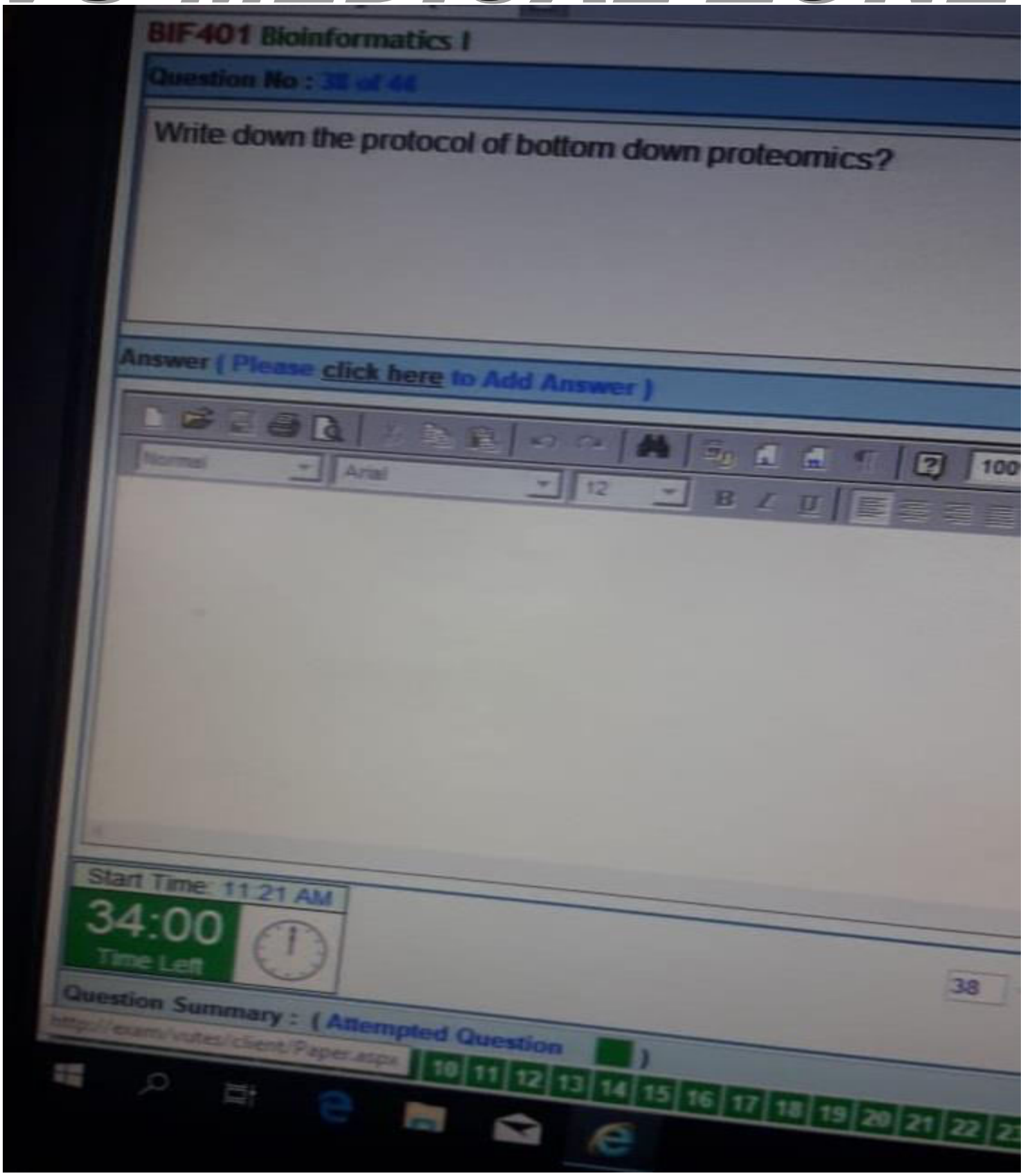
- Consider variable coil and loop sizes besides the from tetra peptide turns

- Consider local protein folding environments
- Solvent accessibility of residues
- Protein structural class
- Protein's organism

Chou Fasman can be improved to better predict secondary structures by incorporating biochemical factors and updated statistics!

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VU MEDICAL ZONE



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VU MEDICAL ZONE

Module 011: TOP DOWN PROTEOMICS

Bottom up proteomics identifies the proteins by cleaving them into segments at specific sites and was not suitable to measure the direct protein masse.

- **PROTOCOL**

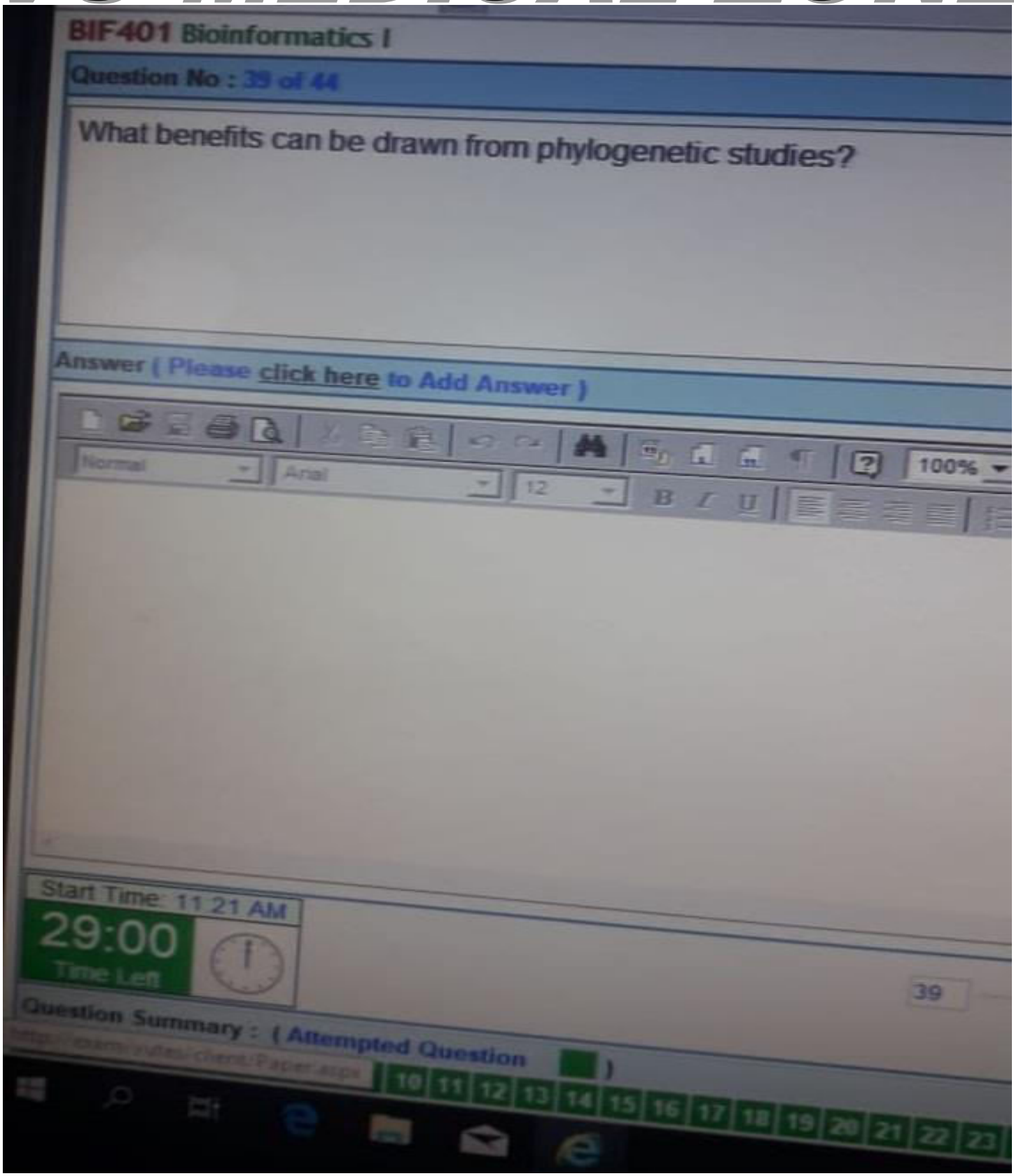
Qno 38

1. Sample containing the protein mixture from cells and tissues is obtained
2. The entire protein is mixed and analyzed for masses.
3. The list of masses is obtained.
4. TDP Measures all post translational masses of protein.
5. After MS1 one protein is selected at a time and fragmented to obtain its peptides.
6. The process is repeated many times.

Comparison is done from protein database uniProt and swissProt.

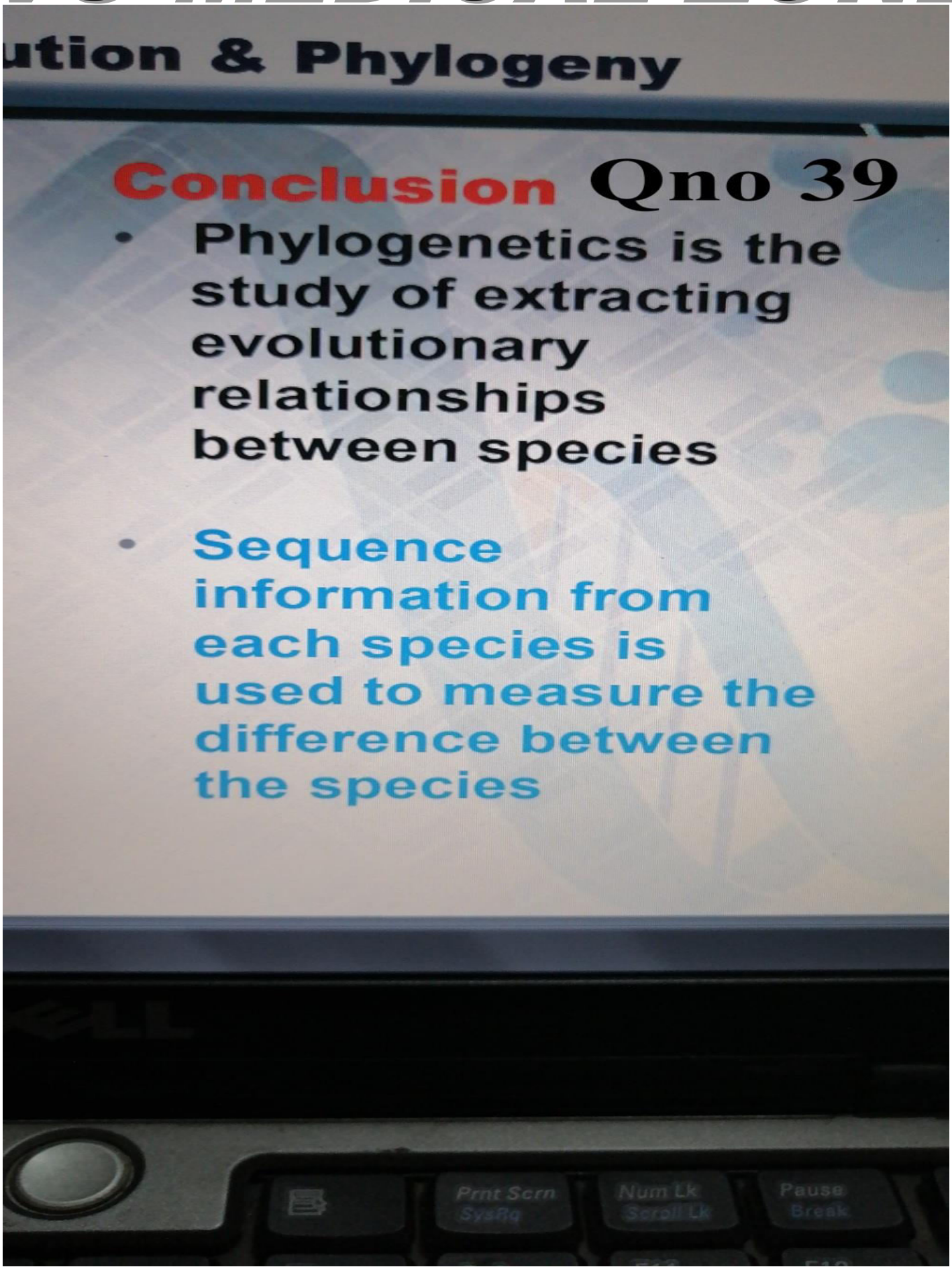
EFFORT BY AMAAN KHAN

VU MEDICAL ZONE



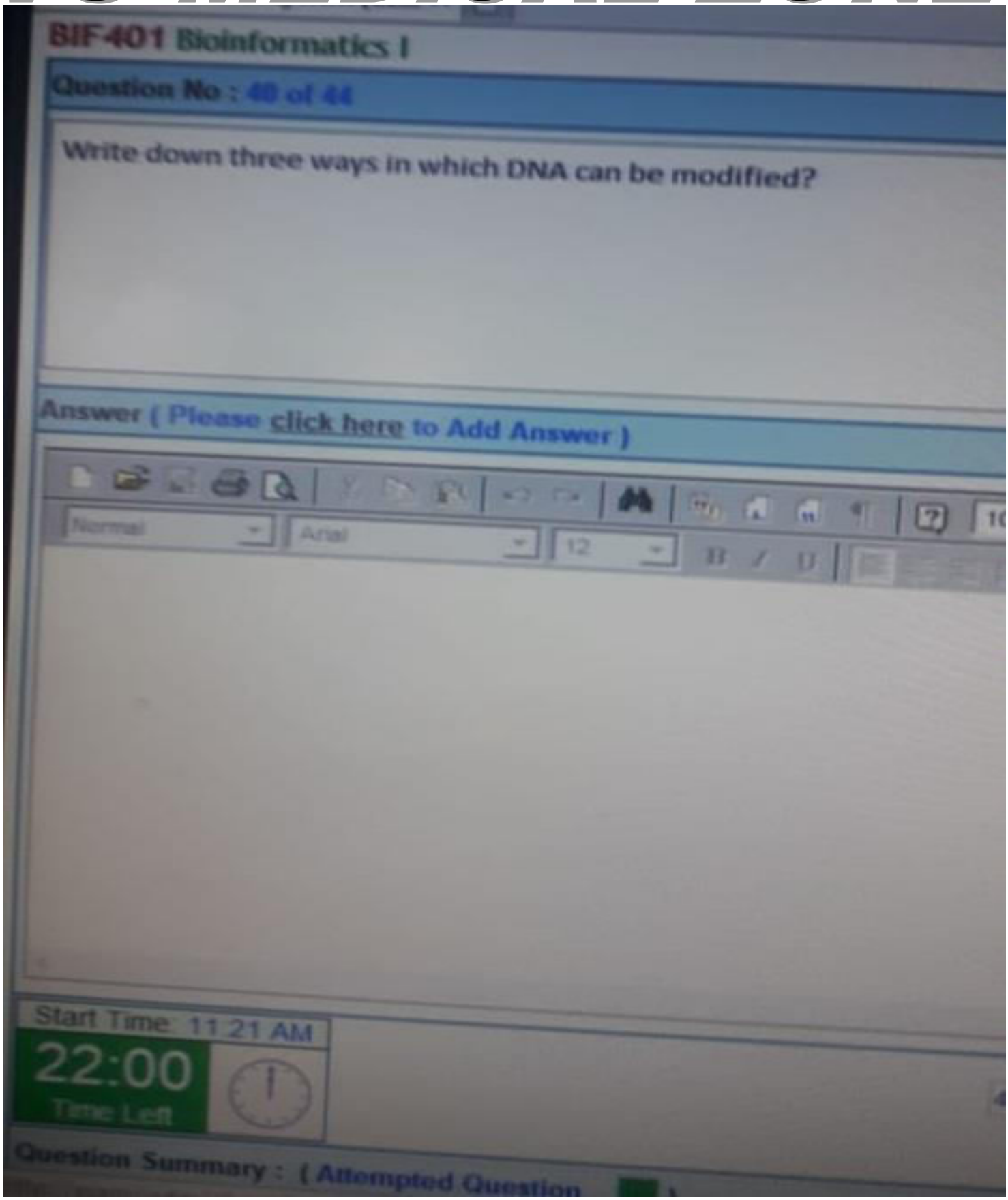
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VU MEDICAL ZONE



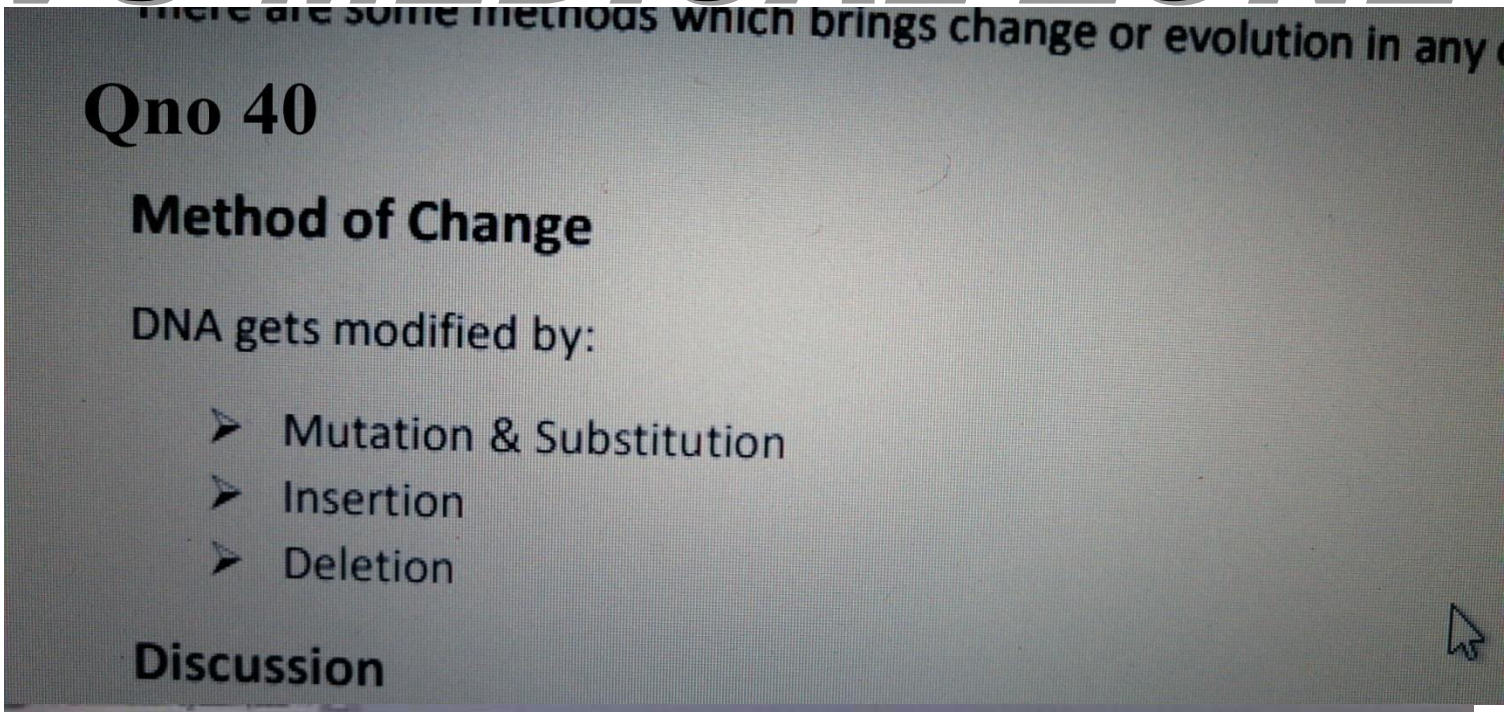
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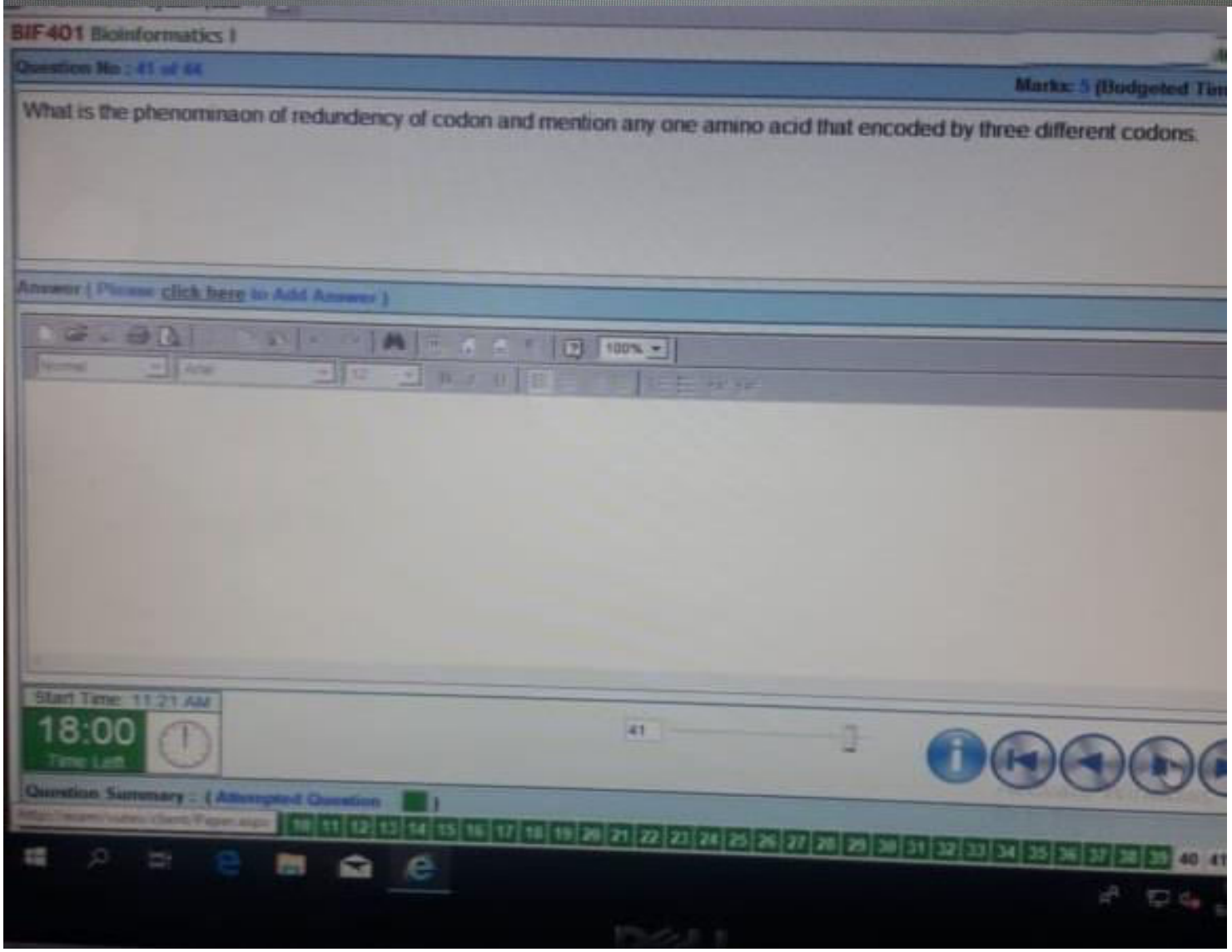
VU MEDICAL ZONE



EFFORT BY AMAAN KHAN

VU MEDICAL ZONE





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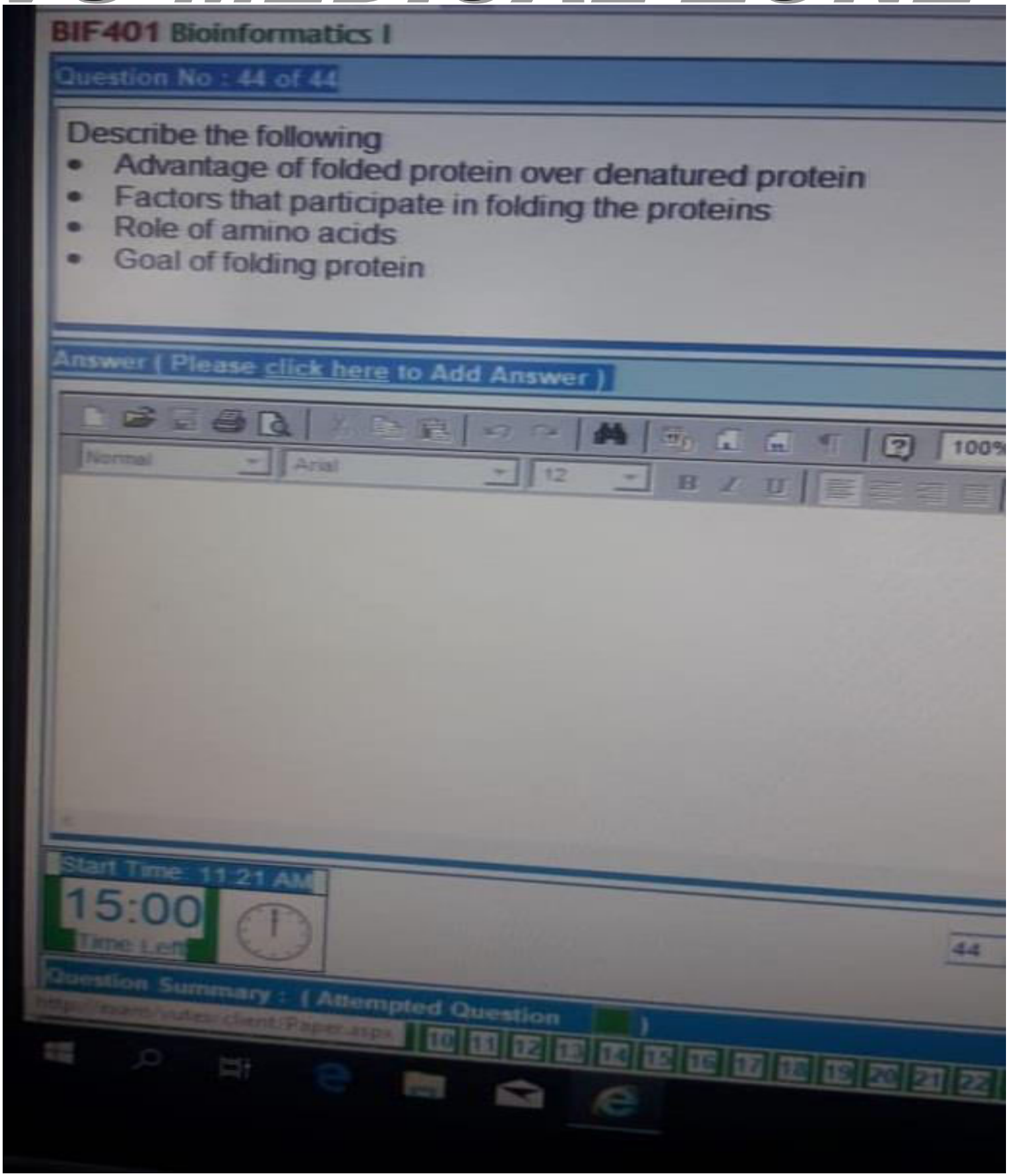
VU MEDICAL ZONE

Qno 41

Redundancy in the genetic code means that most amino acids are specified by more than one mRNA **codon**. For example, the amino acid phenylalanine (Phe) is specified by the **codons** UUU and UUC, and the amino acid leucine (Leu) is specified by the **codons** CUU, CUC, CUA, and CUG.

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https://folding.stanford.edu/home/the-science/

Qno 44

Role of Amino Acids

We know that amino acids can be polar, charged and hydrophobic. Role of polar and charged amino acids in folding. Role of hydrophobic amino acids in folding.

Overall Goal of Folding

Anfinsen's thermodynamic hypothesis: Proteins fold for a unique, stable and minimum free kinetic energy structure. What other factors may come into play for satisfying Anfinsen hypothesis.

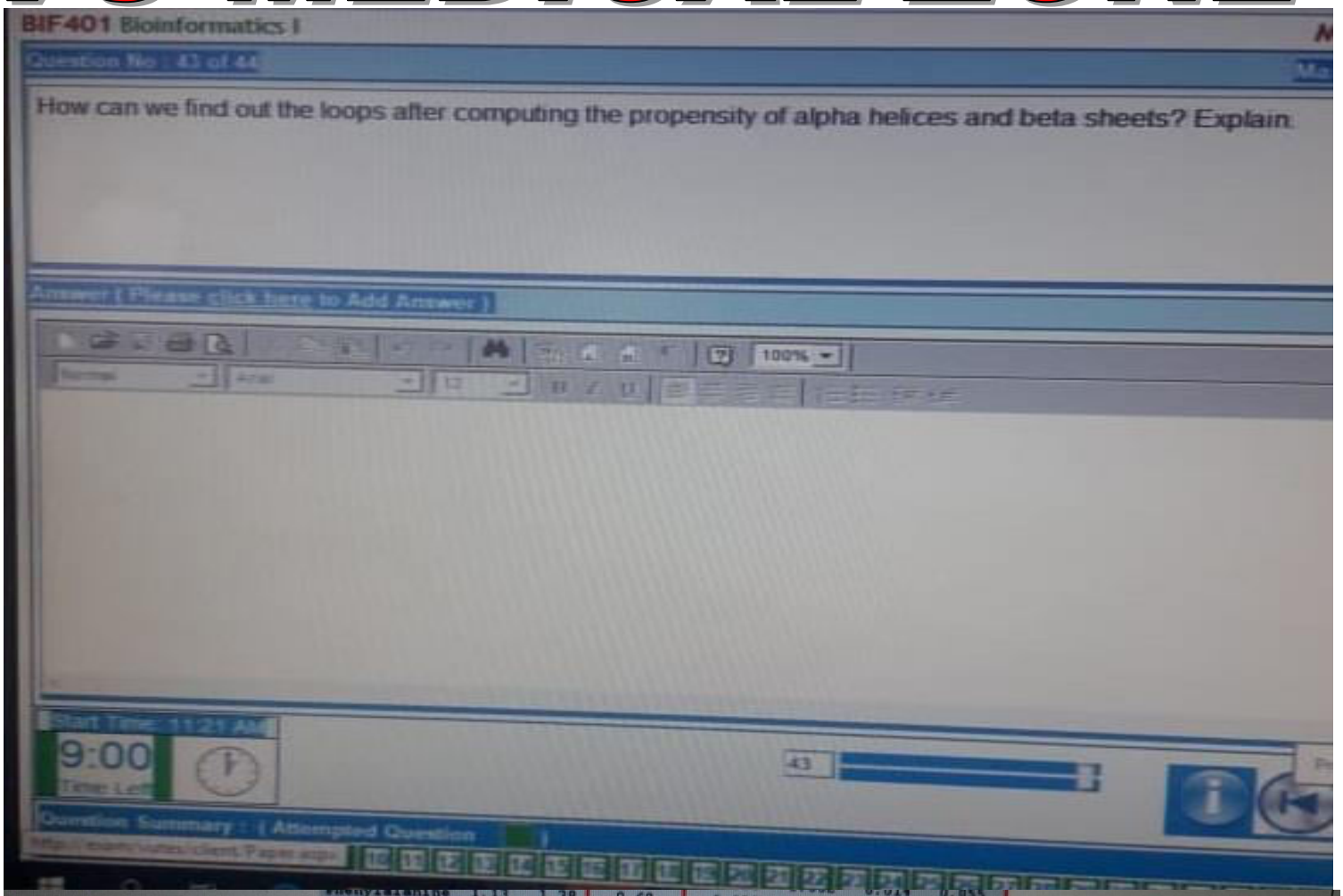
Minimizing Energy

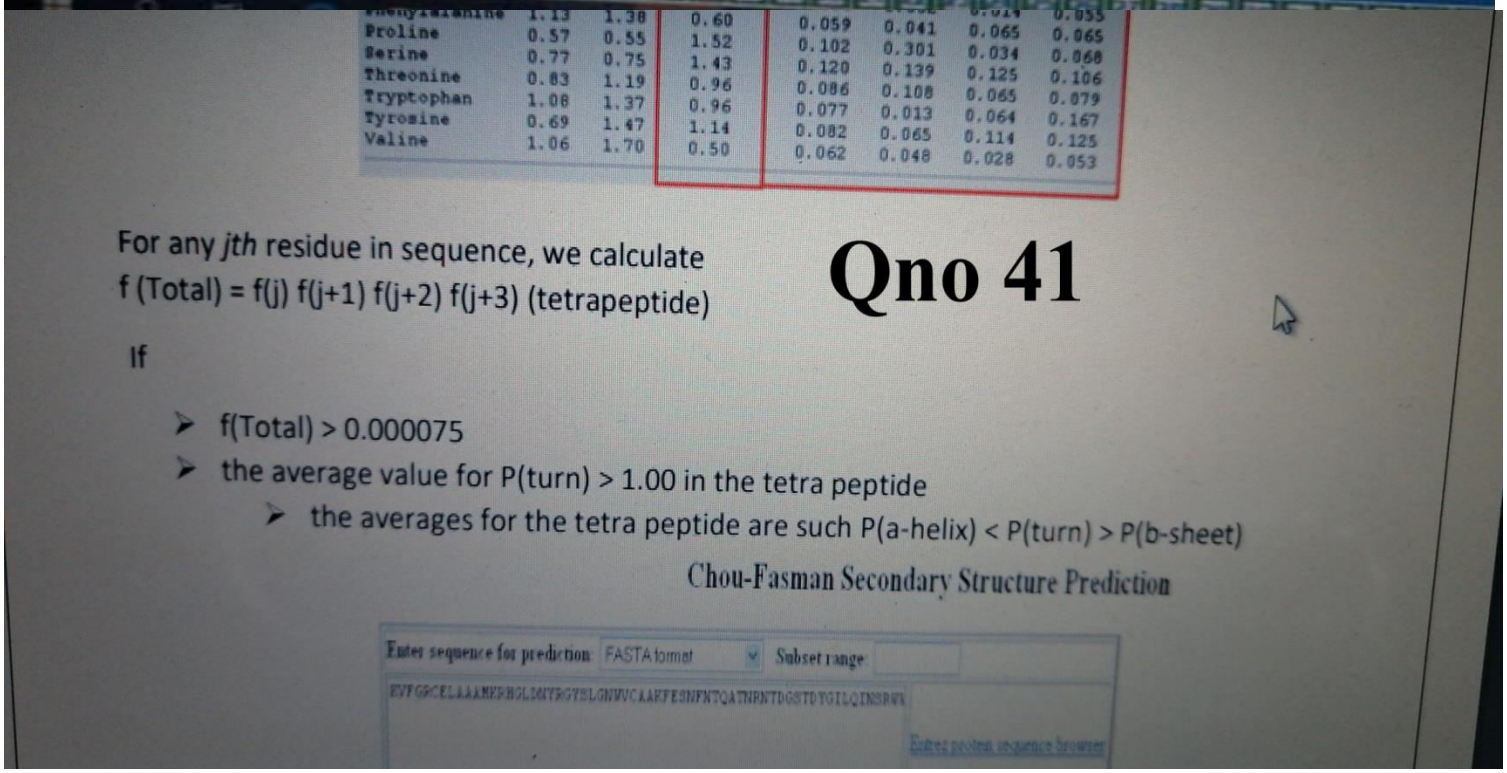
properties of amino acids. Solubility and reactivity are key factors participating in protein folding.

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Introduction to Ab Initio Modelling

Qno 34

Limitation

- Computationally expensive
- Suitable for proteins with less than 100 residues

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